Lipids

Dr. Maha Ali

Learning Objectives

- **Lipids, classification and biomedical importance**
- **Fatty acids, EFA, Trans FA, classification and biomedical importance**
- **Simple lipids, chemistry and properties**
- **Phospholipids, classification and functions**

Continued:

- Eicosanoids, classification and importance
- Lipoproteins, classification and significance
- Cholesterol, structure and functions
- Bile acids/salts and functions
- Lipid peroxidation & its significance
- Obesity, and CAD(Coronary artery disease.

Lipids

Lipids are biological molecules(Plant or Animal) that are insoluble in water but soluble in organic solvents (ether, chloroform, benzene, acetone etc).

Or

Lipids are naturally occurring molecule from a plant or animal soluble in non polar organic solvents.

Lipids

- **Definition:**
- Lipids are heterogeneous group of compounds which are soluble in organic or non-polar solvents, like ether, chloroform, benzene, acetone etc.
- **Examples:** fats, oil, steroid, waxes, phospholipids etc.
- **Chemically:** Lipids are generally esters of fatty acids with alcohol

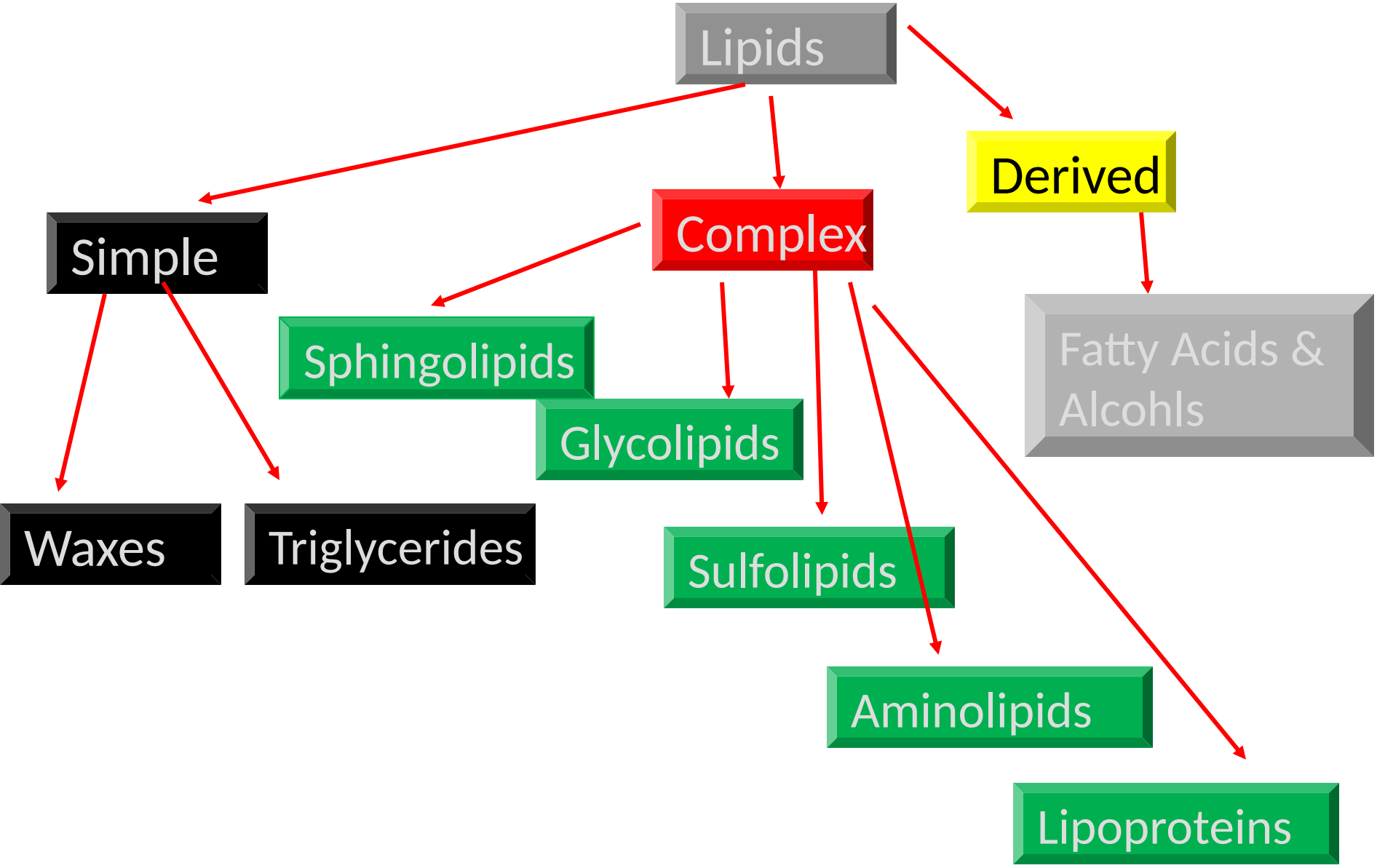
Foods Rich in Lipids

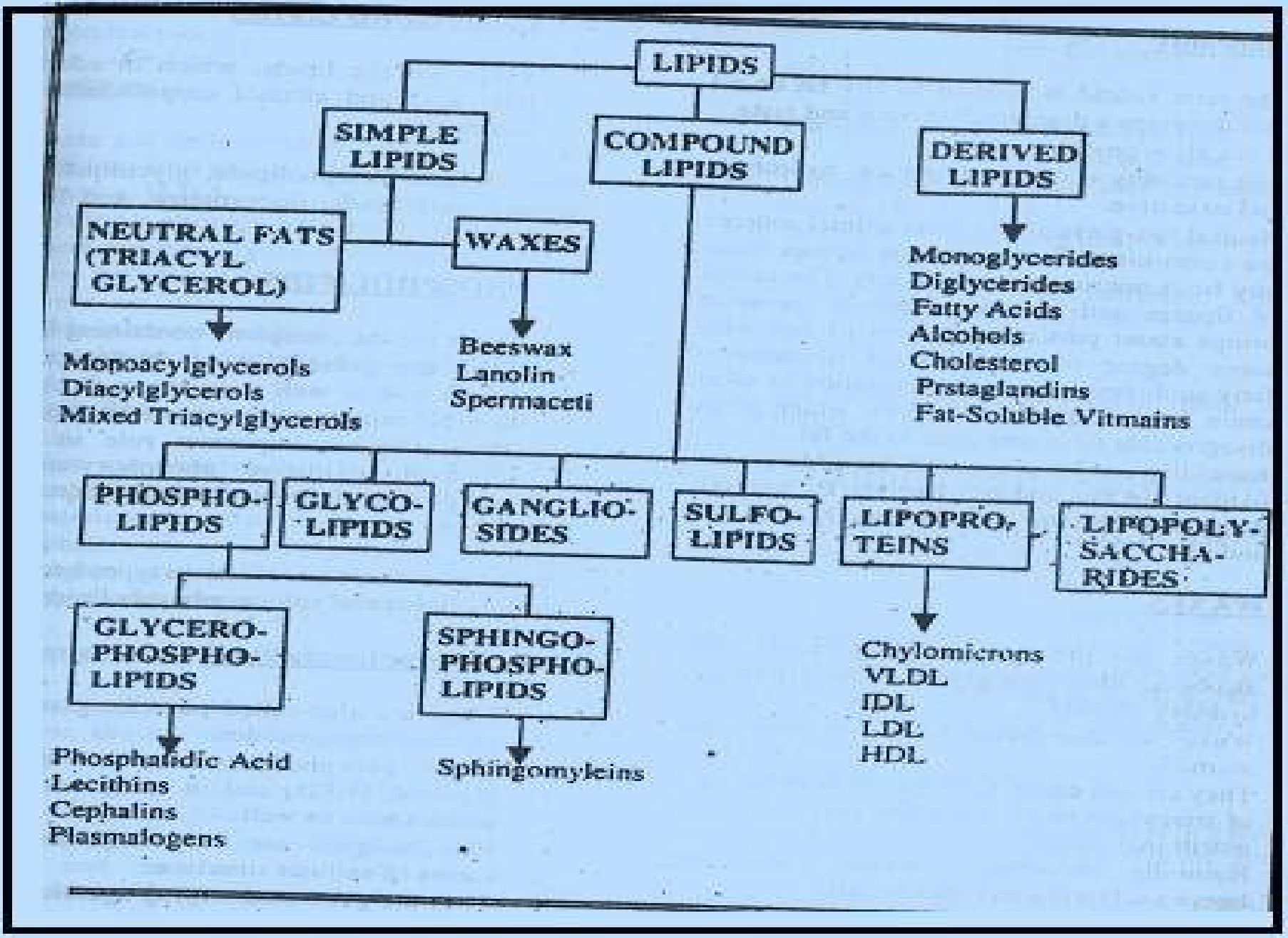


Biomedical Importance of Lipids

- An energy source, lipids provide 9 kcal of energy per gram(CHO 4 kcal/gram).
- Triglycerides serve as energy storage in adipocytes
- Phosphoglycerides, sphingolipids, and steroids are structural components of cell membranes.
- Steroid hormones (Aldosterone, cortisol, estrogen & testosterone etc) are essential for different body functions.
- Lipid-soluble vitamins (A, E, D, K). Dietary fat acts as a carrier of lipid-soluble vitamins into cells of small intestine
- Provide shock absorption(around the organs) and insulation
- Dipalmitoylecithin (**dipalmitoylphosphatidylcholine) (DPPC),** is a component of lung surfactant.

Lipid Classification




LIPIDS
SIMPLE LIPIDS
COMPOUND LIPIDS
DERIVED LIPIDS
NEUTRAL FATS (TRIACYL GLYCEROL)
WAXES
Monoacylglycerols
Diacylglycerols
Mixed Triacylglycerols
Beeswax
Lanolin
Spermaceti
Monoglycerides
Diglycerides
Fatty Acids
Alcohols
Cholesterol
Prstaglandins
Fat-Soluble Vitmains
PHOSPHO-LIPIDS
GLYCO-LIPIDS
GANGLIO-SIDES
SULFO-LIPIDS
LIPOPRO-TEINS
LIPOPOLY-SACCHA-RIDES
GLYCERO-PHOSPHO-LIPIDS
SPHINGO-PHOSPHO-LIPIDS
Phosphatidic Acid
Lecithins
Cephalins
Plasmalogens
Sphingomyleins
Chylomicrons
VLDL
IDL
LDL
HDL

Classification of Lipids

- Simple lipids. Ester of Fatty Acids with alcohols.
  - Neutral fats(Triglycerides)= FA+ Glycerol
  - Waxes. FA + Monohydroxy alcohols
    - True Waxes. FA+ Cetyle alcohol(C_{13} H_{33} OH)
    - Cholesterol ester
    - Vitamins A & D esters
- Compound Lipids. Fatty Acids+ Alcohols+ Other groups
    - Phospholipids
    - Glyco lipids
    - Sulpho lipids
    - Amino lipids
    - Lipoproteins(Chylomicron, VLDL, **IDL(intermediate-density lipoproteins** (**IDL**) accumulate in your blood. **IDL** has a cholesterol-to-triglycerides ratio that's higher than that for VLDL. This disorder results in high plasma levels of both cholesterol and triglycerides. Type 4 is a dominantly inherited) ,LDL &HDL)
- Derived Lipids. Derived from simple & compound
  - Fatty Acids
  - Cholesterol
  - PGs
  - Fat soluble vitamins

Classification of Lipids on the basis of saponification

Based on saponfication. It is the formation of soap.

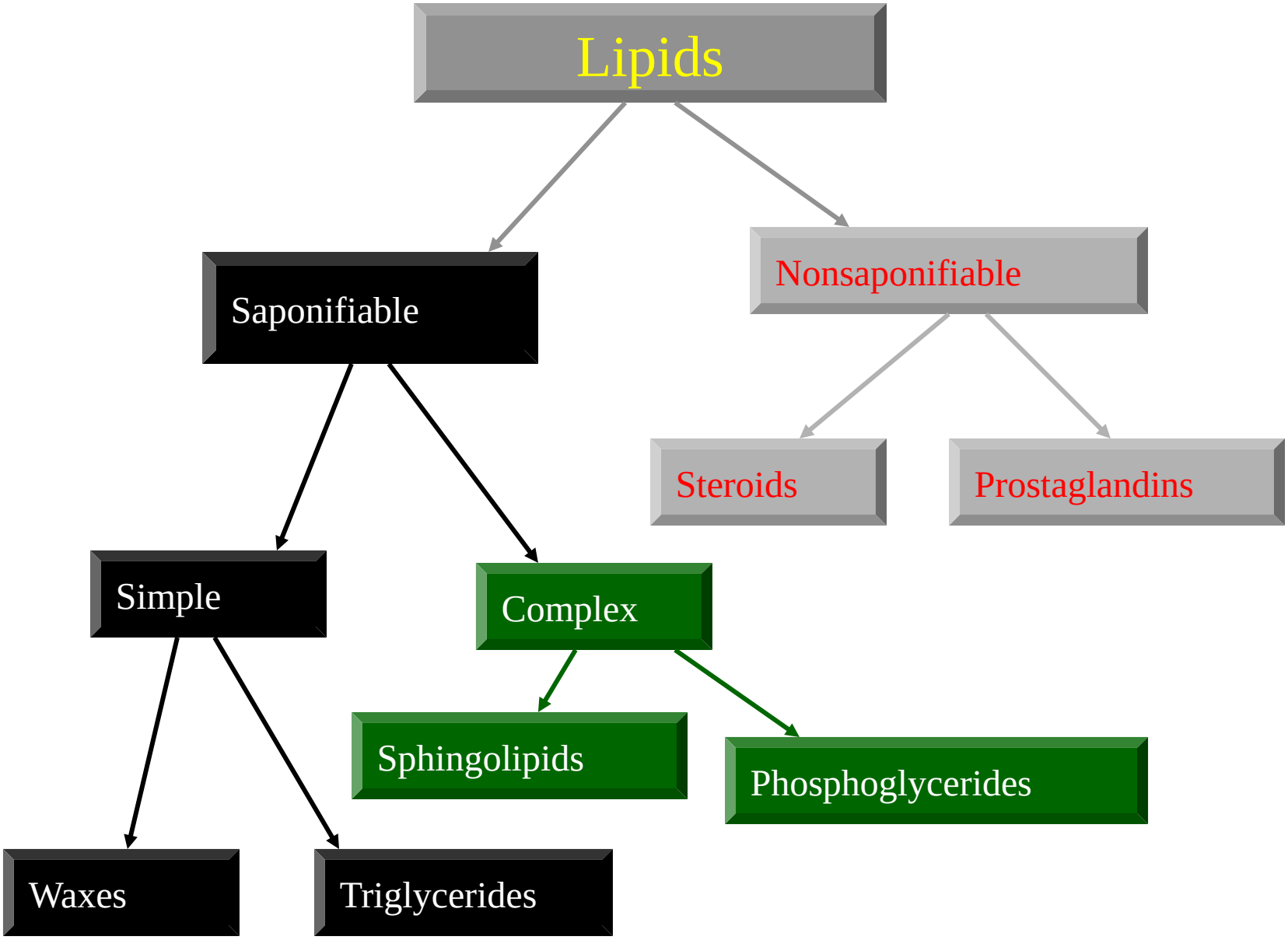
When lipids(fatty acids) are boiled with Alkali (NaOH &KOH) form soaps. lipids have two subclasses

Saponifiable

- **Simple:** has two sub-categories
  - Waxes
  - Triglycerides
- **Complex**: has two sub-categories
  - Phosphoglycerides
  - Sphingolipids

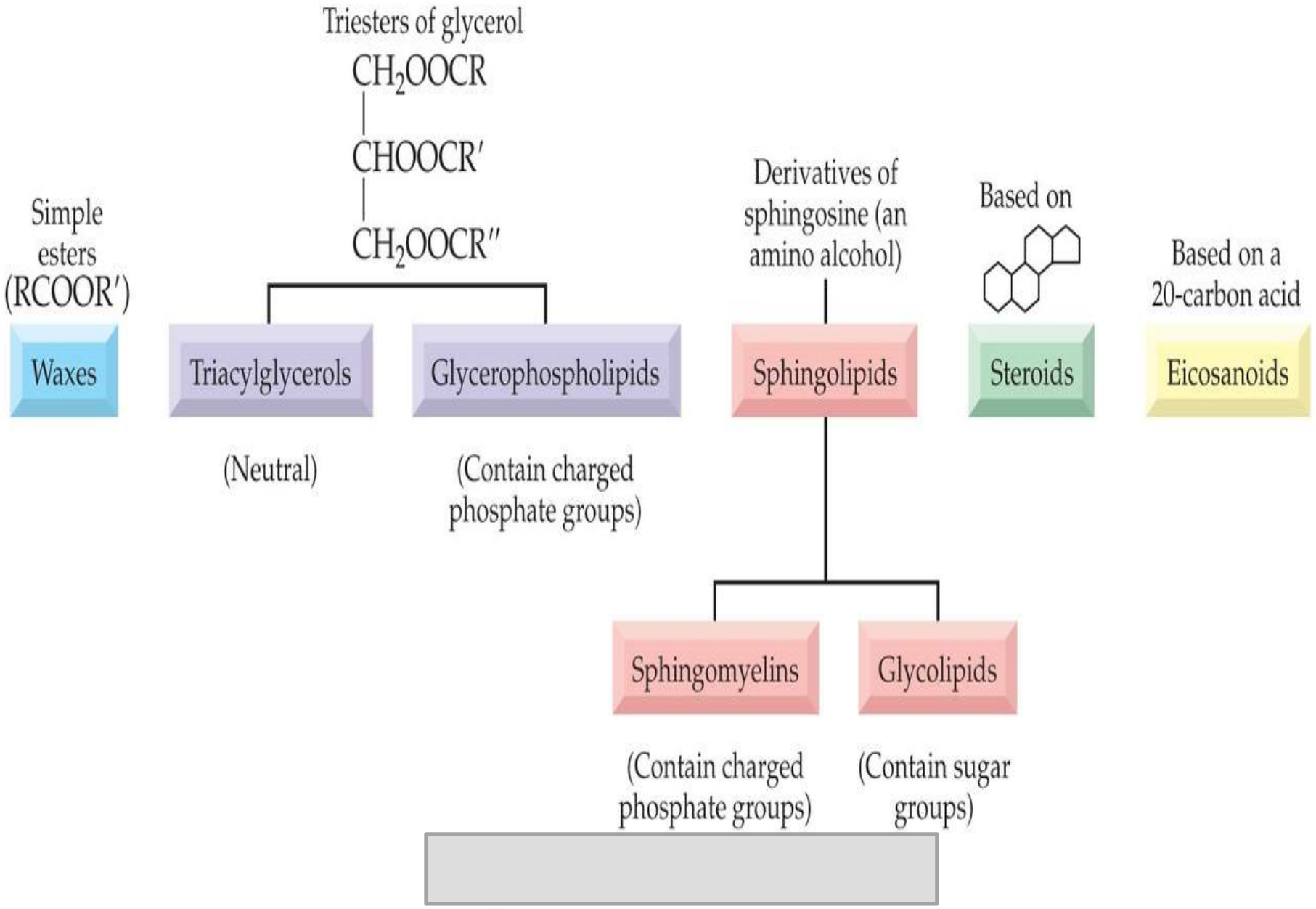
- **Nonsaponifiable** have two subclasses
  - Steroids
  - Prostaglandins

Classification of Lipids on the basis of saponification



KEY POINTS

- **Lipid:** A naturally occurring molecule from a plant or animals, soluble in nonpolar organic solvents.
- **Fatty acid:** A long-chain carboxylic acid; those in animal fats and vegetable oils often have 12–22 carbon atoms.
- **Waxes** are carboxylic acid esters, RCOOR’,with long, straight hydrocarbon chains in both R groups; they are secreted by sebaceous glands in the skin of animals and perform mostly external protective functions.


Triesters of glycerol
CH2OOCR
CHOOCR′
CH2OOCR″
Simple esters (RCOOR′)
Waxes
Triacylglycerols
(Neutral)
Glycerophospholipids
(Contain charged phosphate groups)
Derivatives of sphingosine (an amino alcohol)
Sphingolipids
Based on
Steroids
Based on a 20-carbon acid
Eicosanoids
Sphingomyelins
(Contain charged phosphate groups)
Glycolipids
(Contain sugar groups)

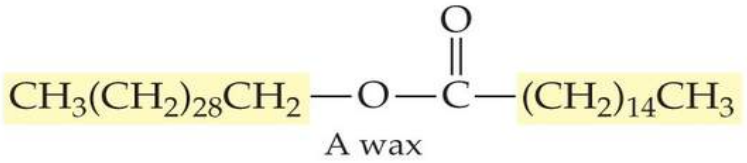
- **Glycerophospholipids.**

  Are triesters of glycerol.Together with other lipids, they help to control the flow of molecules into and out of cells.
- **Sphingomyelins.**

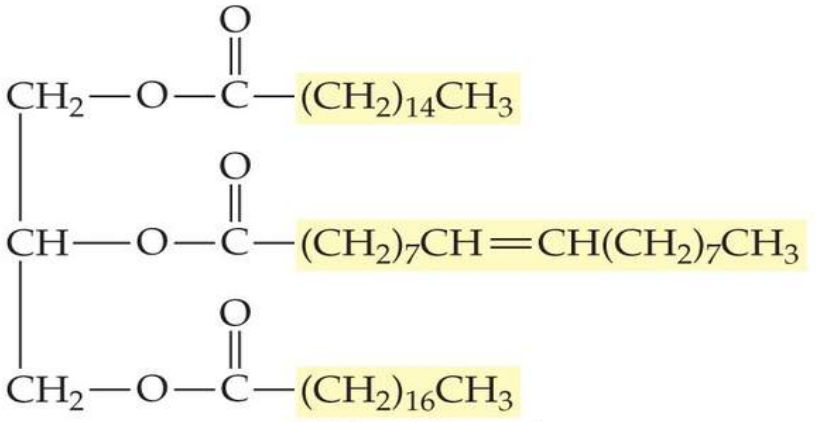
  Amides derived from an amino alcohol (sphingosine)they are essential to the structure of cell membranes and are abundant in nerve cell membranes.
- **Glycolipids.**

  CHO containing amides derived from sphingosine, contain polar carbohydrate groups; on cell surfaces the carbohydrate portion is recognized by and connects to intracellular messengers.

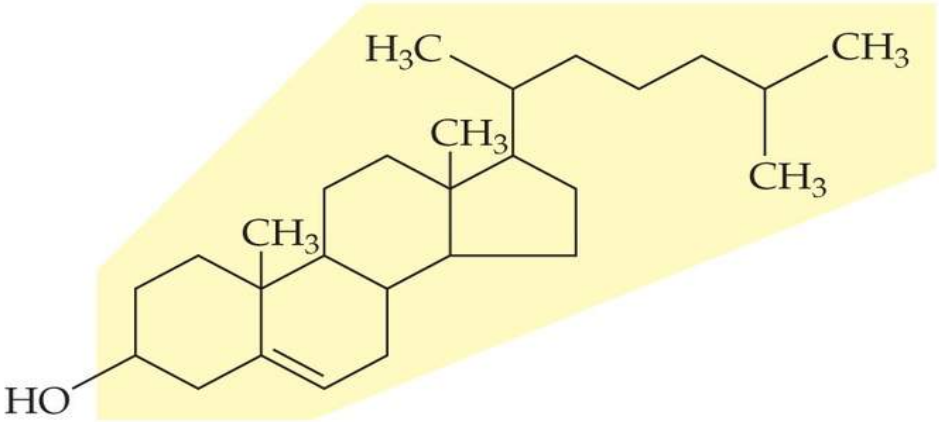
Non esters or non amides groups of lipids the steroids and the eicosanoids.



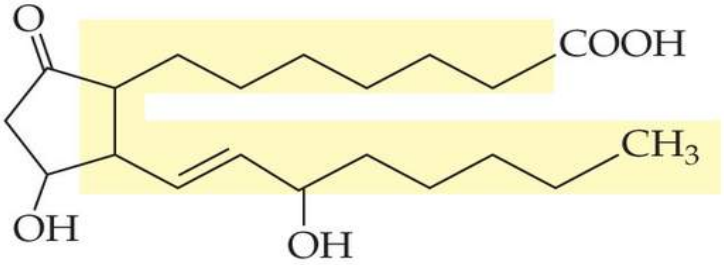
A wax



A triacylglycerol



Cholesterol, a steroid



A prostaglandin

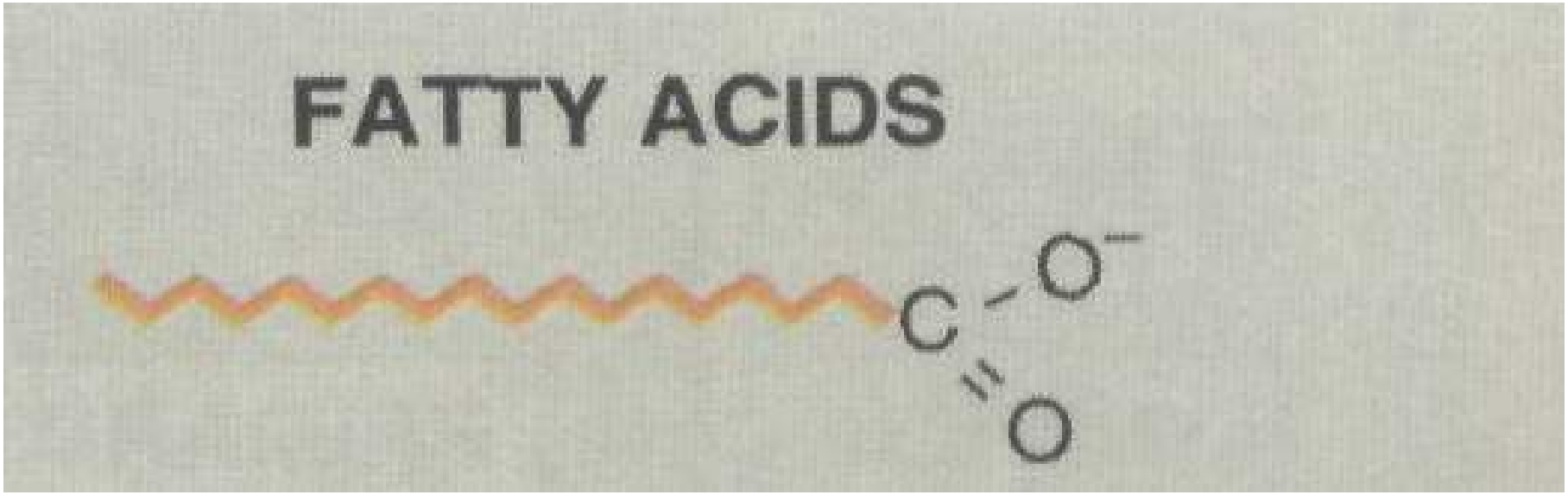
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Simple and Complex Lipids

- Simple: an ester-containing lipid with just two types of components
  - An alcohol
  - One or more fatty acids
- Complex: an ester-containing lipid with more than two components
  - An alcohol
  - Fatty acids
  - Plus others

FattyAcids

- Fatty acids are the derived lipids containing a long hydrocarbon chain and a terminal carboxylate group.
- They are the aliphatic mono carboxylic organic acids made up of hydrocarbon chains.



Fatty acids vary in chain length and degree of unsaturation:

- Usually contain an even number of carbon atoms, typically between 4 and 24.
- The 16- and 18-carbon fatty acids are most common.

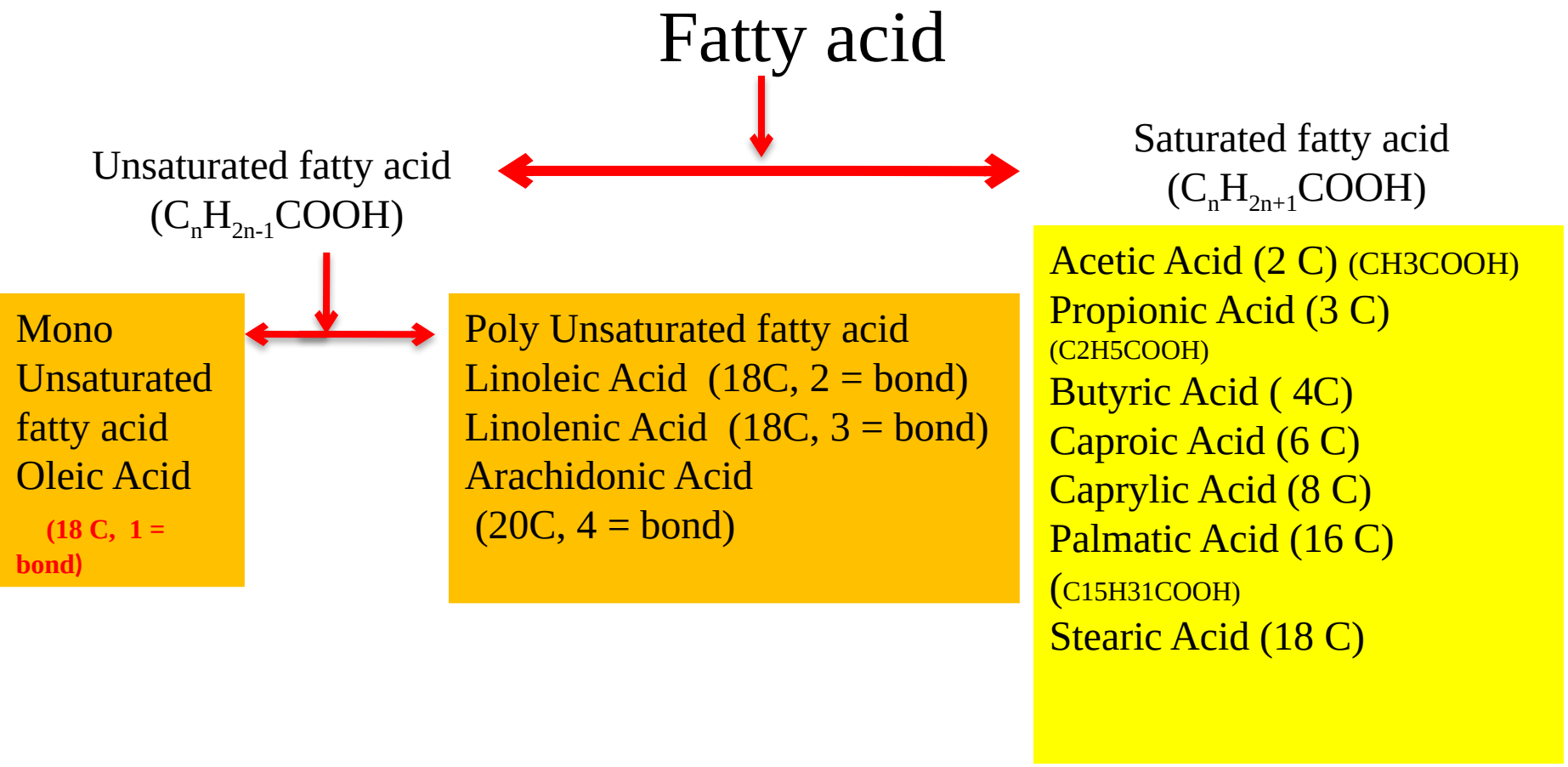
- May contain one or more double bonds.

Classification of Fatty Acid

- Depending up on the presences or absences of a double bond

- Fatty acid are classified on the basis of saturation and un-saturation.

Classification of Fatty Acid

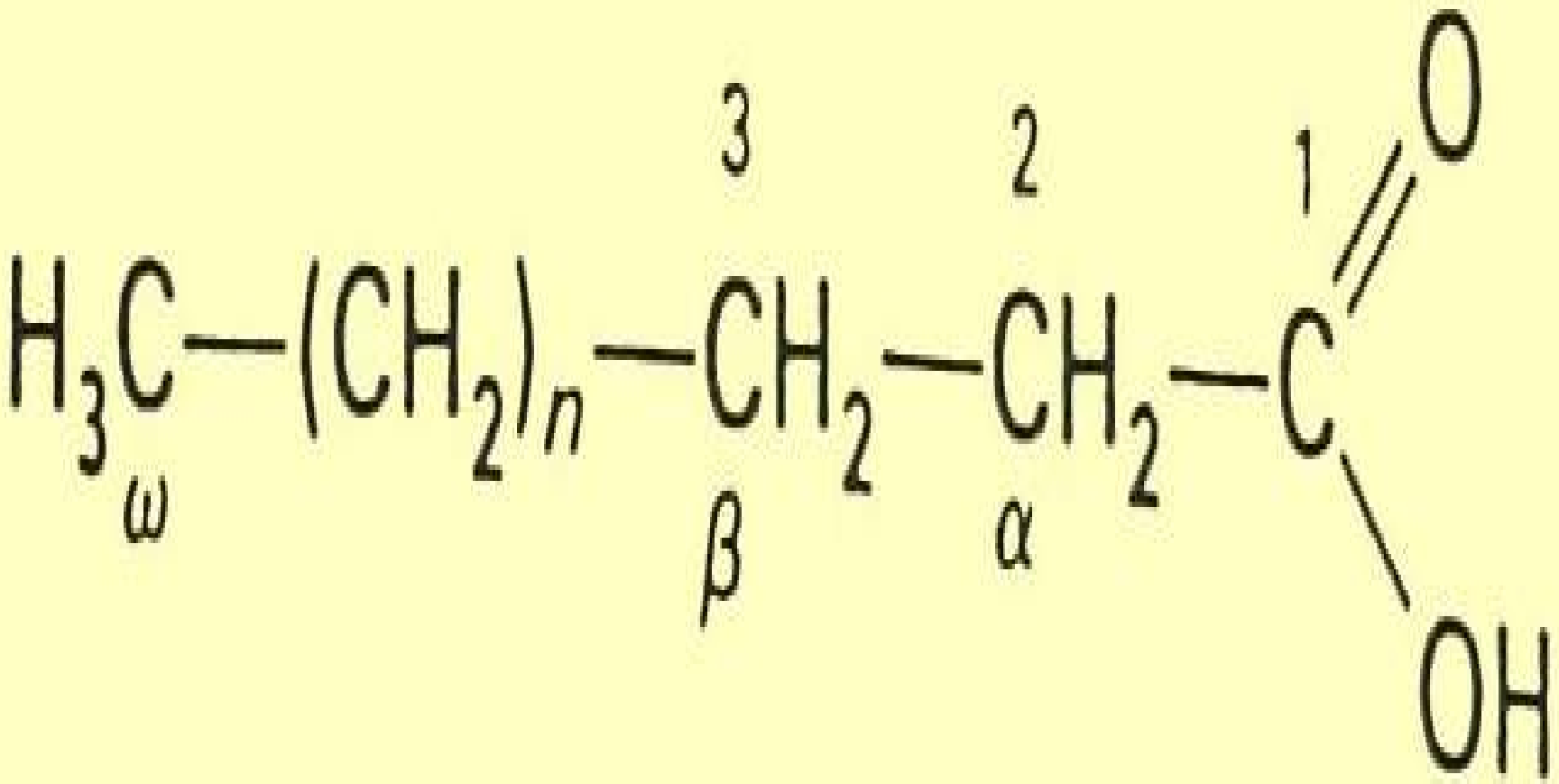


Classification of Fatty Acid on the basis of Chain length

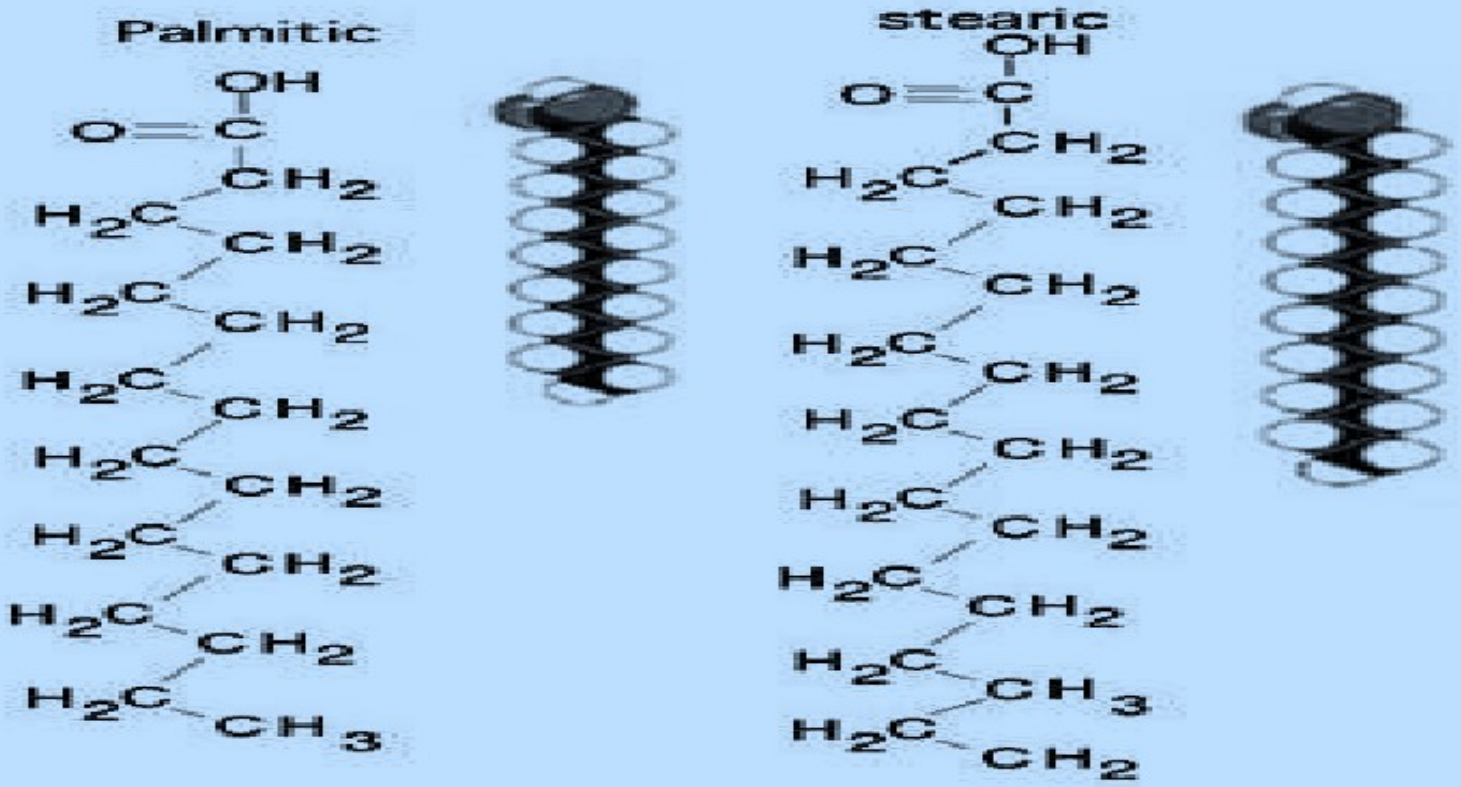
On the basis of Chain length: fatty acids may be

- Short chain FA. 4-6 carbons
- Medium chain. 8-12 carbons:
- Long chain. 14+ Carbons:

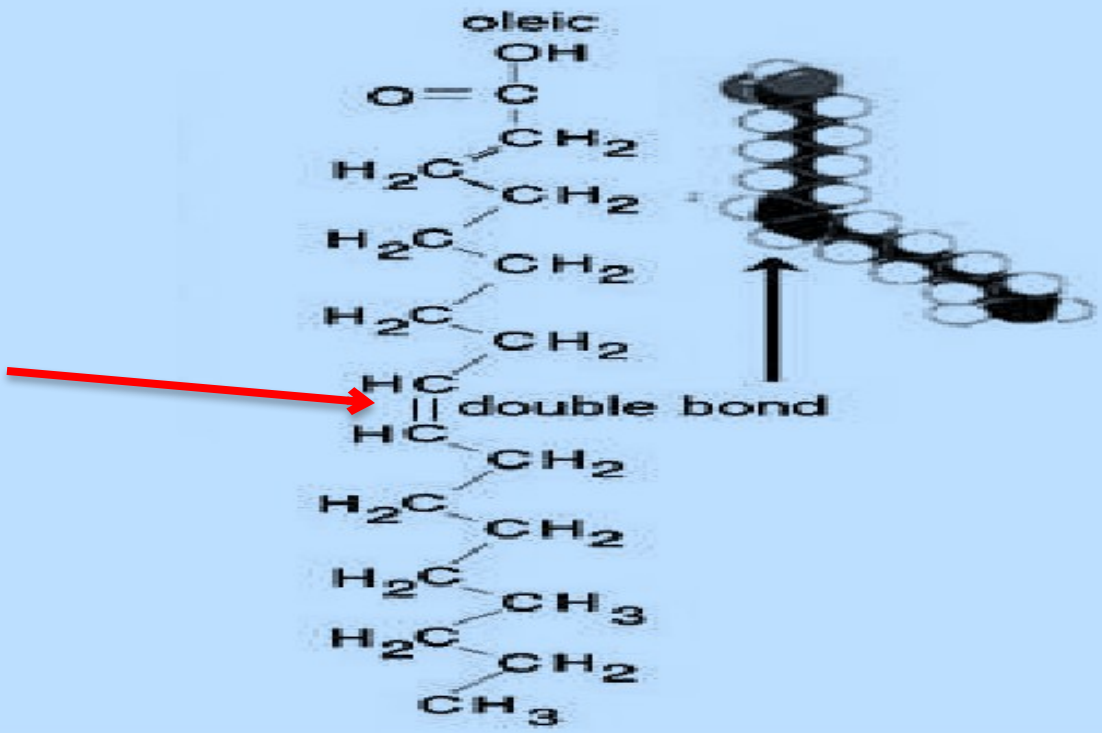
F.A. Nomenclature



Saturated Fatty Acid (SFA)

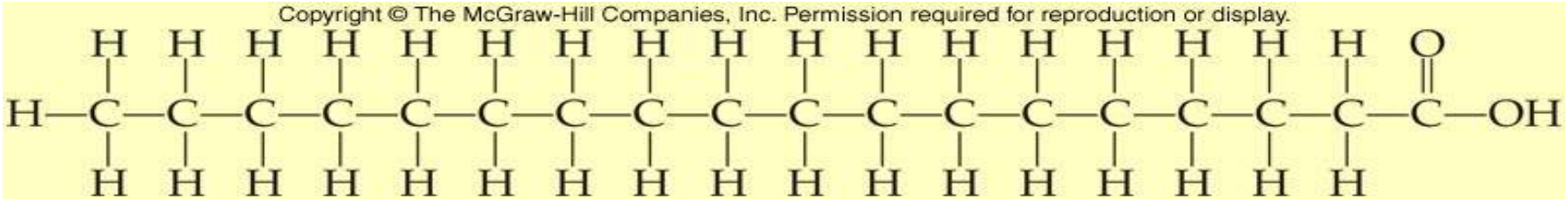


MUFA's

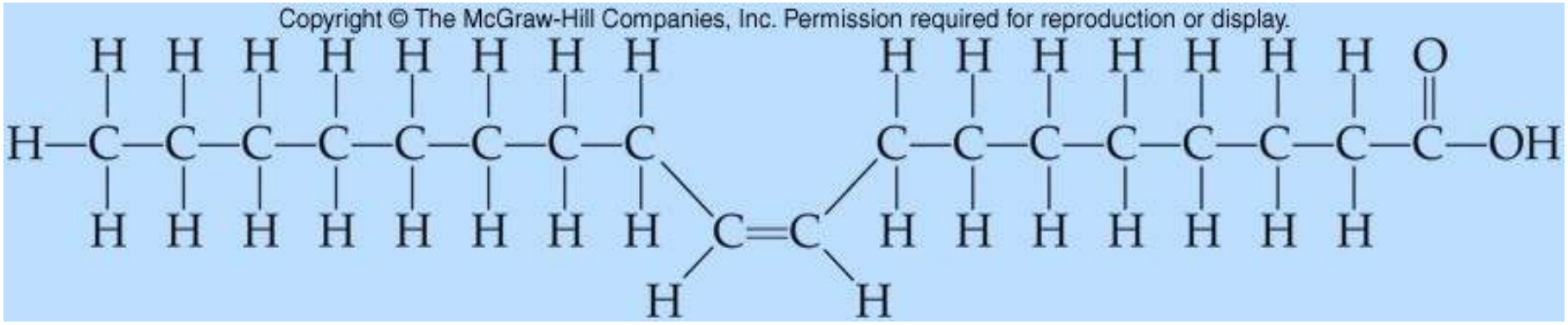

oleic
OH
O
C
CH2
H2C
CH2
H2C
CH2
H2C
CH2
H2C
CH2
HC
double bond
HC
CH2
H2C
CH2
H2C
CH3
H2C
CH2
CH3

FA

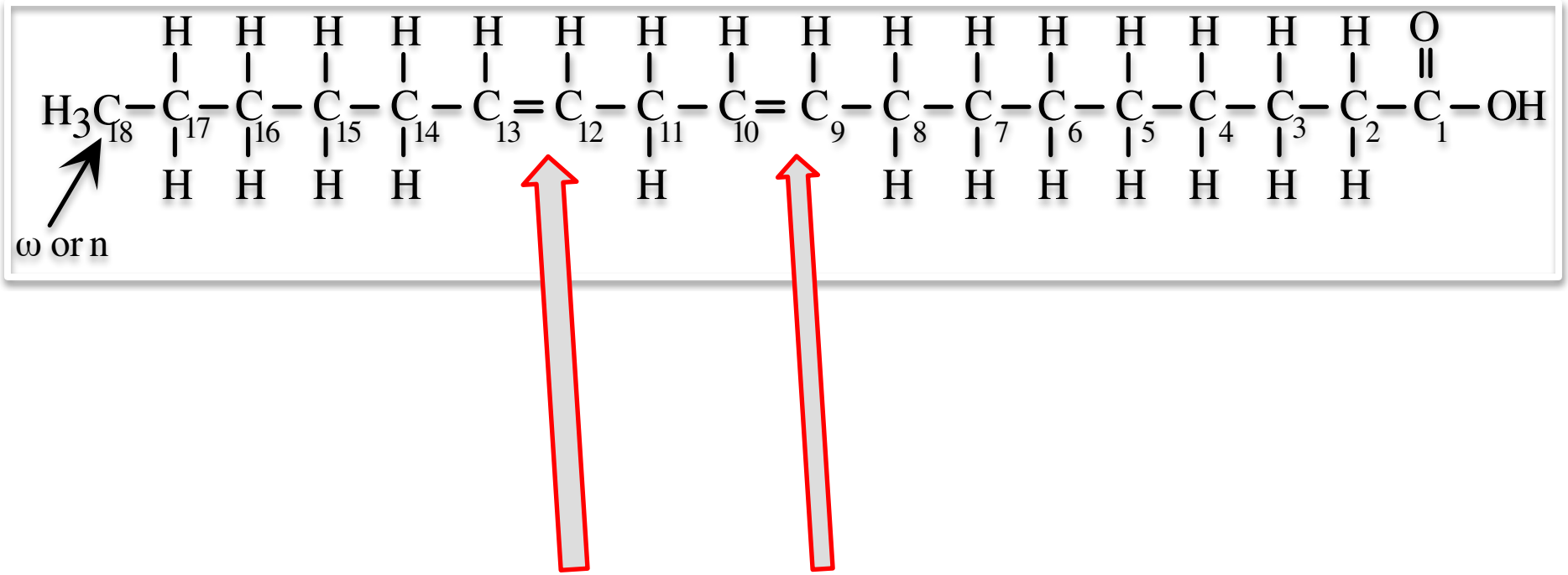
- Stearic acid: a typical saturated fatty acid with 18 carbons in the chain



- Oleic acid: a typical unsaturated fatty acid with 18 carbons in the chain(MUFA)



PUFA's



Saturated & Unsaturated Fatty Acids

Common Saturated Fatty Acids

Common Name	I.U.P.A.C. Name	Melting Point (°C)	RCOOH Formula	Condensed Formula
Capric	Decanoic	32	$C_9H_{19}COOH$	$CH_3(CH_2)_8COOH$
Lauric	Dodecanoic	44	$C_{11}H_{23}COOH$	$CH_3(CH_2)_{10}COOH$
Myristic	Tetradecanoic	54	$C_{13}H_{27}COOH$	$CH_3(CH_2)_{12}COOH$
Palmitic	Hexadecanoic	63	$C_{15}H_{31}COOH$	$CH_3(CH_2)_{14}COOH$
Stearic	Octadecanoic	70	$C_{17}H_{35}COOH$	$CH_3(CH_2)_{16}COOH$
Arachidic	Eicosanoic	77	$C_{19}H_{39}COOH$	$CH_3(CH_2)_{18}COOH$

Common Unsaturated Fatty Acids

Common Name	I.U.P.A.C. Name	Melting Point (°C)	RCOOH Formula	Number of Double Bonds	Position of Double Bonds
Palmitoleic	*cis*-9-Hexadecenoic	0	$C_{15}H_{29}COOH$	1	9
Oleic	*cis*-9-Octadecenoic	16	$C_{17}H_{33}COOH$	1	9
Linoleic	*cis,cis*-9,12-Octadecadienoic	5	$C_{17}H_{31}COOH$	2	9, 12
Linolenic	All *cis*-9,12,15-Octadecatrienoic	−11	$C_{17}H_{29}COOH$	3	9, 12, 15
Arachidonic	All *cis*-5,8,11,14-Eicosatetraenoic	−50	$C_{19}H_{31}COOH$	4	5, 8, 11, 14

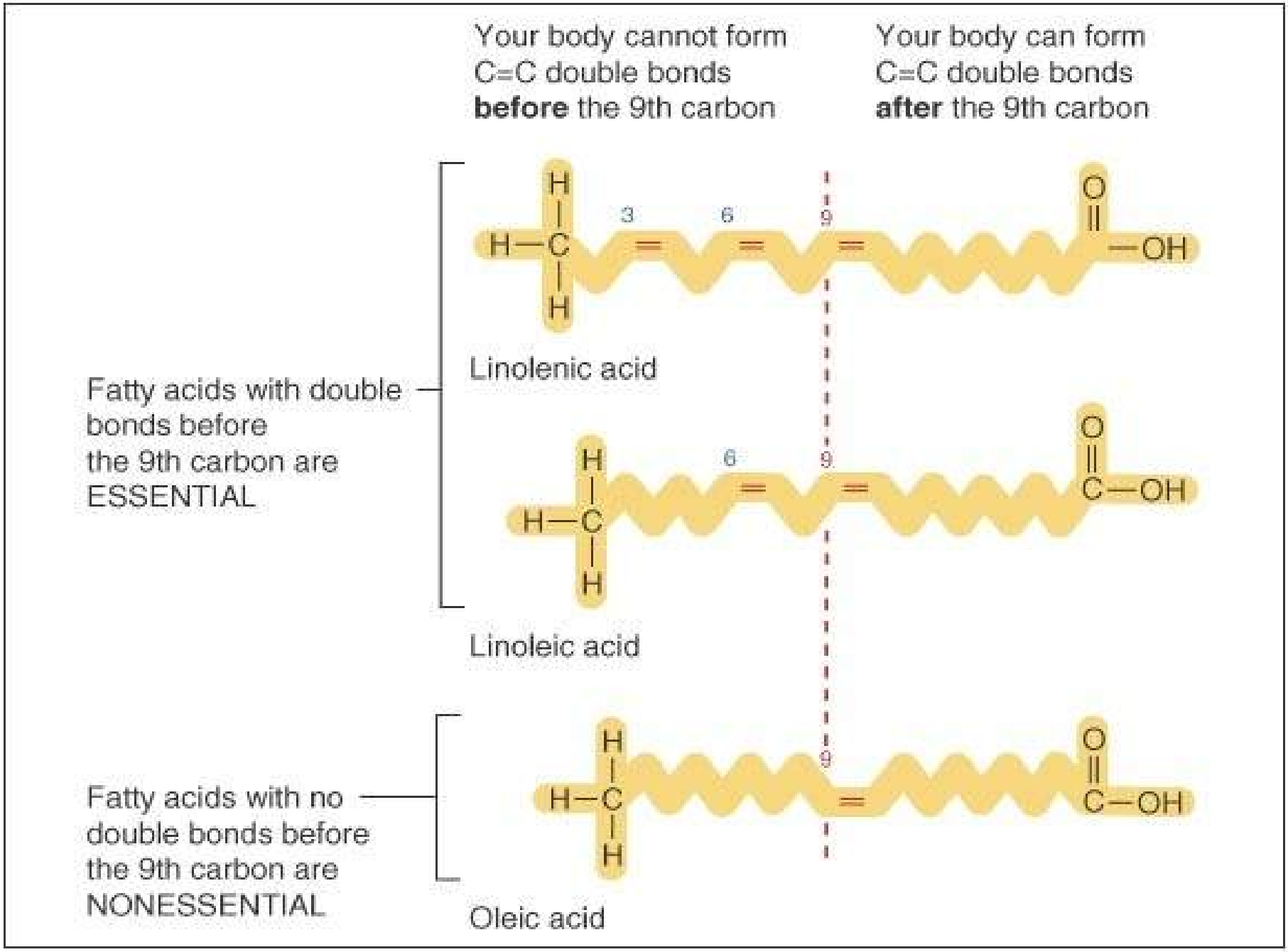
Condensed Formula

Palmitoleic	$CH_3(CH_2)_5CH{=}CH(CH_2)_7COOH$
Oleic	$CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH$
Linoleic	$CH_3(CH_2)_4CH{=}CH{-}CH_2{-}CH{=}CH(CH_2)_7COOH$
Linolenic	$CH_3CH_2CH{=}CH{-}CH_2{-}CH{=}CH{-}CH_2{-}CH{=}CH(CH_2)_7COOH$
Arachidonic	$CH_3(CH_2)_4CH{=}CH{-}CH_2{-}CH{=}CH{-}CH_2{-}CH{=}CH{-}CH_2{-}CH{=}CH{-}(CH_2)_3COOH$

14:0/18:2

- The notation 14:0 means 14 carbons and 0 double bonds in the molecule (saturated).

- The notation 18:2 means 18 carbons in the molecule and 2 double bonds.


Your body cannot form C=C double bonds **before** the 9th carbon
Your body can form C=C double bonds **after** the 9th carbon
Fatty acids with double bonds before the 9th carbon are ESSENTIAL
Linolenic acid
Linoleic acid
Fatty acids with no double bonds before the 9th carbon are NONESSENTIAL
Oleic acid

Fatty Acid Properties

- Melting point increases with increasing carbon number
- Melting point of a saturated fatty acid is higher than an unsaturated fatty acid with the same number of carbons
- Typical saturated fatty acids are tightly packed together
- ***cis*** double bonds prevent good alignment of molecules in unsaturated fatty acids leading to poor packing
- Double bonds lower melting point relative to saturated acid

Q

How would the melting point of stearic acid compare to the melting points of oleic acid and linoleic acid?

Assign the melting points of –17°C, 13°C, and 69°C to the correct fatty acid. Explain.

Stearic acid (18 C) saturated

Oleic acid (18 C) one double bond

Linoleic acid (18 C) two double bonds

A

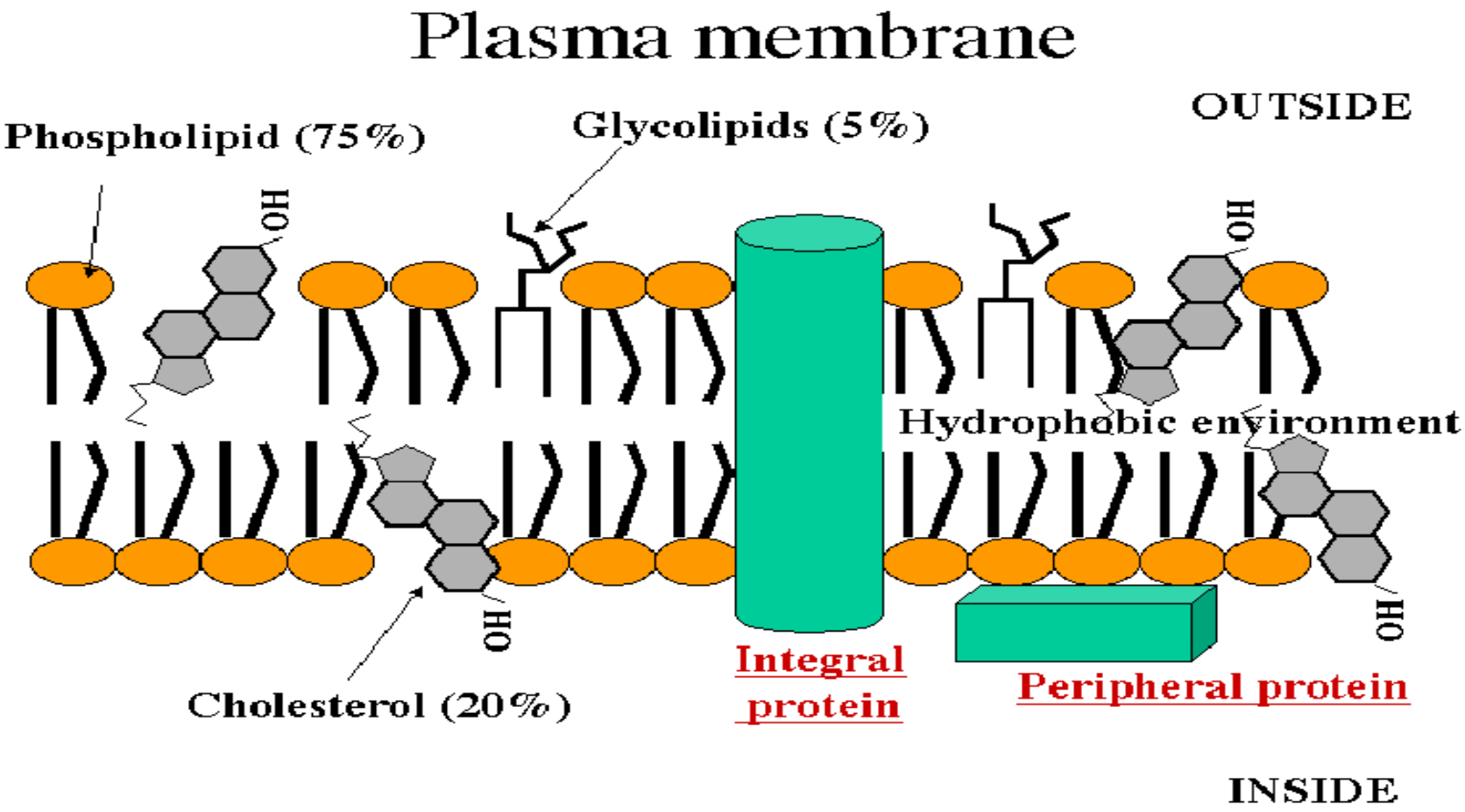
Stearic acid is saturated and would have a higher melting point than the unsaturated fatty acids. Because linoleic has two double bonds, it would have a lower mp than oleic acid, which has one double bond.

Stearic acid mp 69°C

Oleic acid mp 13°C

Linoleic acid mp -17°C

Functions of Fatty acids

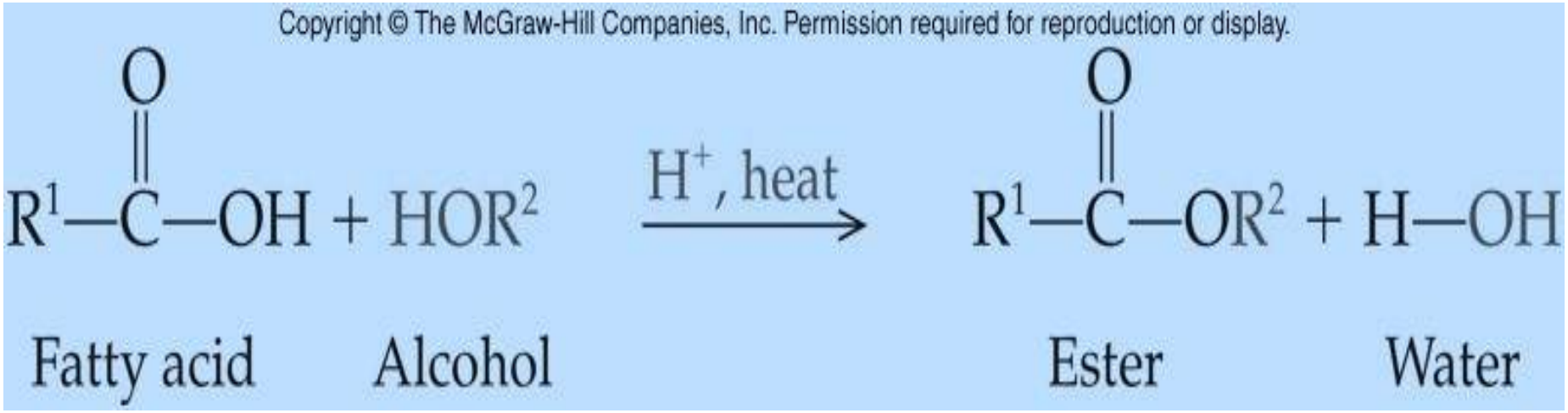


Chemical Reactions of Fatty Acids

1. Esterification of Fatty Acids

Esterification. Fatty acids reacts with alcohols to form esters and water

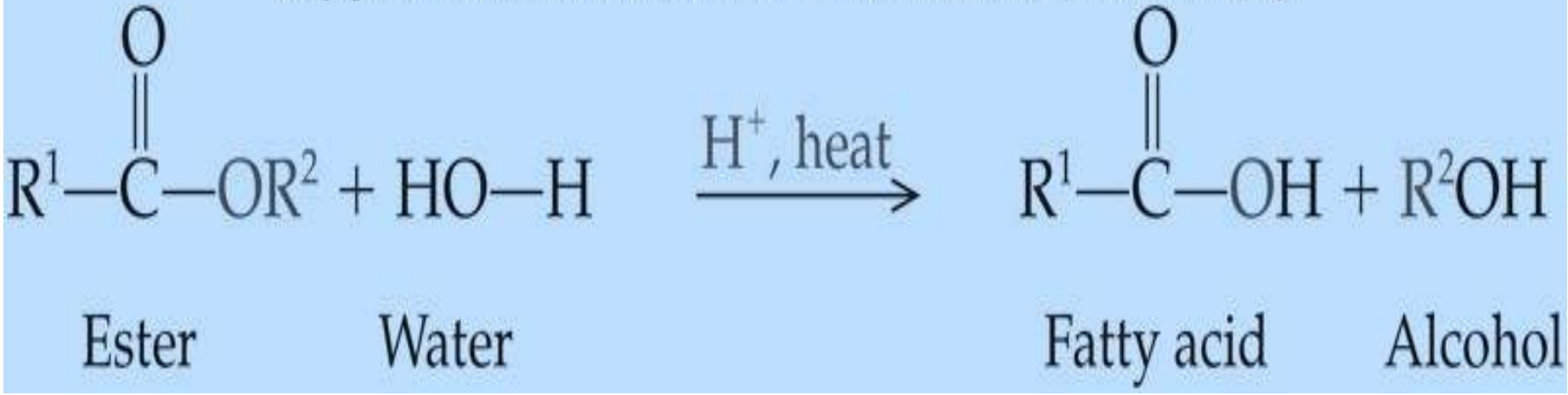
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2. Fatty Acid Hydrolysis

- Acid Hydrolysis reverses esterification
  - Fatty acids are produced from esters

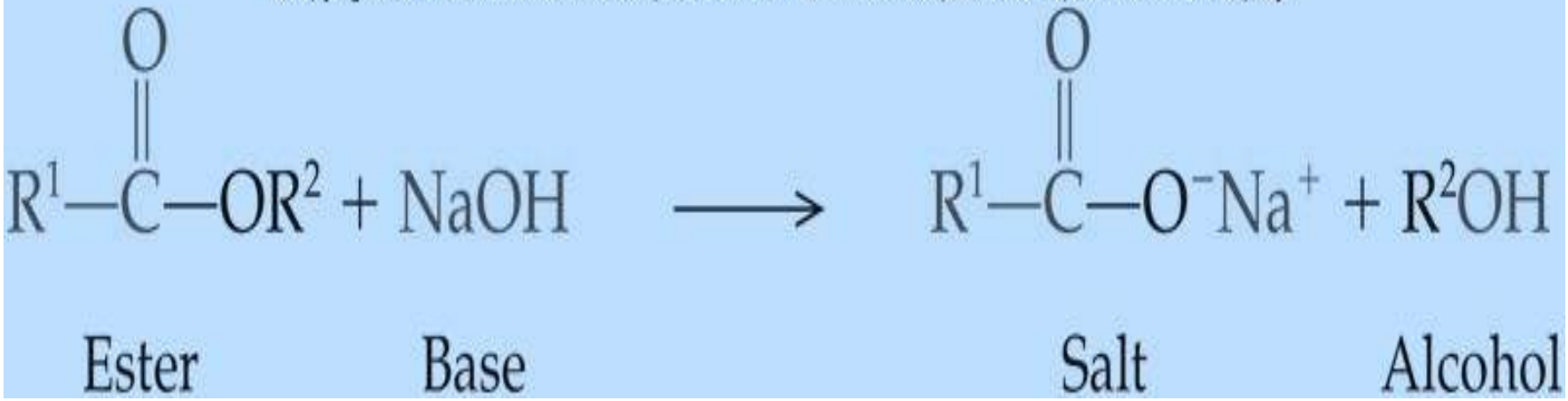
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3. Saponification

- Saponification is the base-catalyzed hydrolysis of an ester
- Products of the reaction are
  - An alcohol
  - An ionized salt which is a soap
    - Soaps have a long uncharged hydrocarbon tail
    - Also have a negatively charged carboxylate group at end
    - Form micelles that dissolve oil and dirt particles

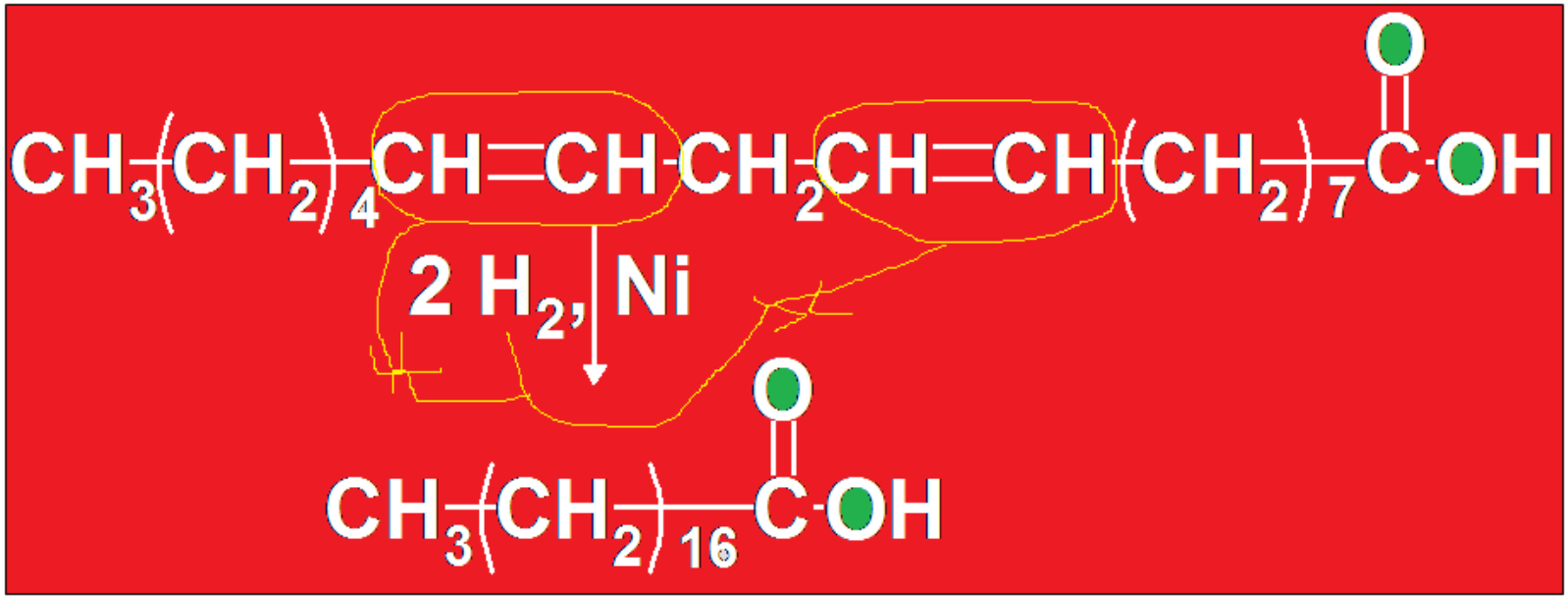
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4. Hydrogenation

Reaction at the Double Bond

- Hydrogenation is an addition reaction
- Unsaturated fatty acids can be converted to saturated fatty acids
- Hydrogenation is used in the food industry

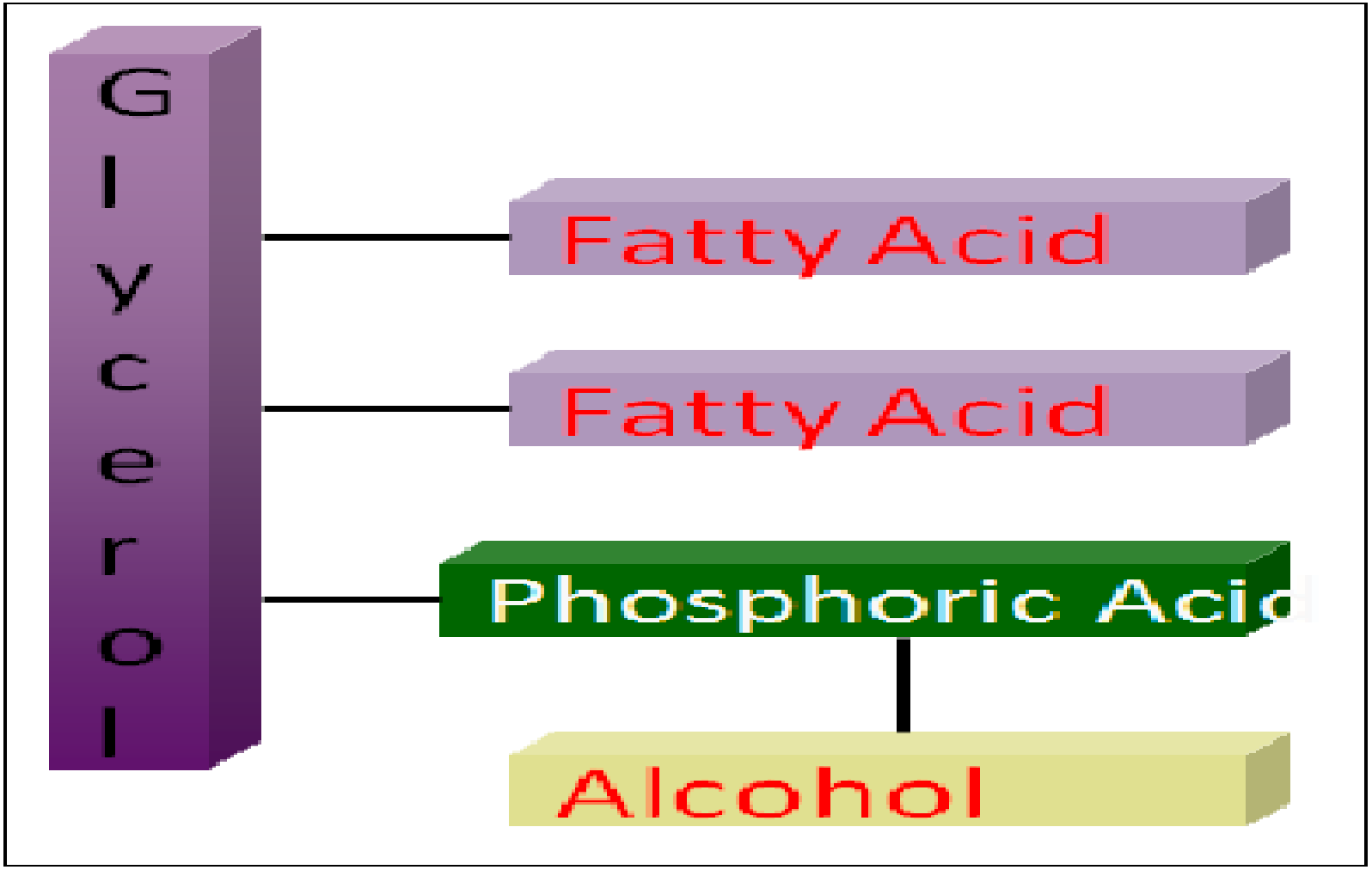


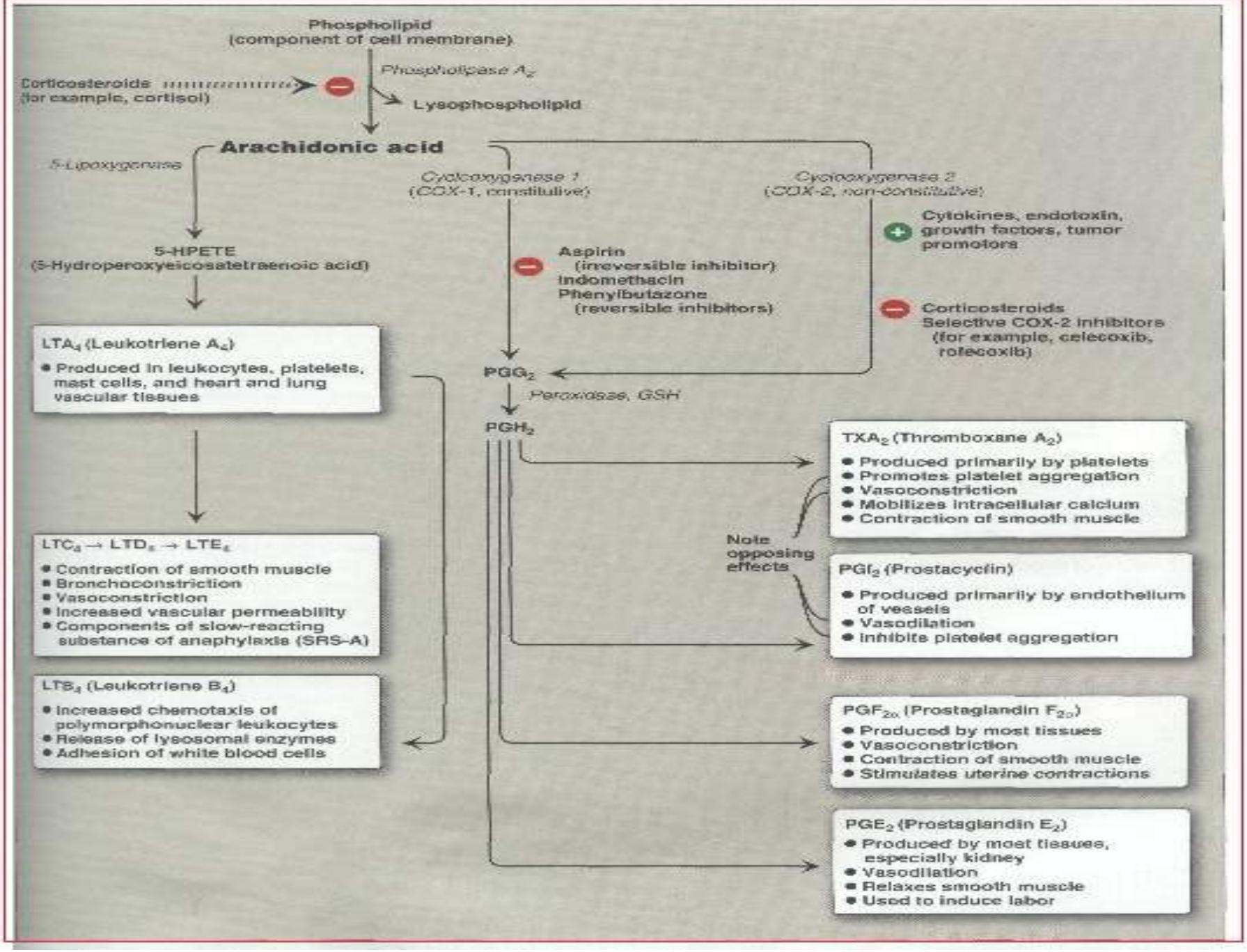
- Eicosanoids: Prostaglandins, Leukotrienes, and Thromboxanes
- Glycerides
- Nonglyceride Lipids
  - Sphingolipids
  - Steroids
- Waxes
- Complex Lipids

Membrane Lipids
Phospholipids &Prostaglandins

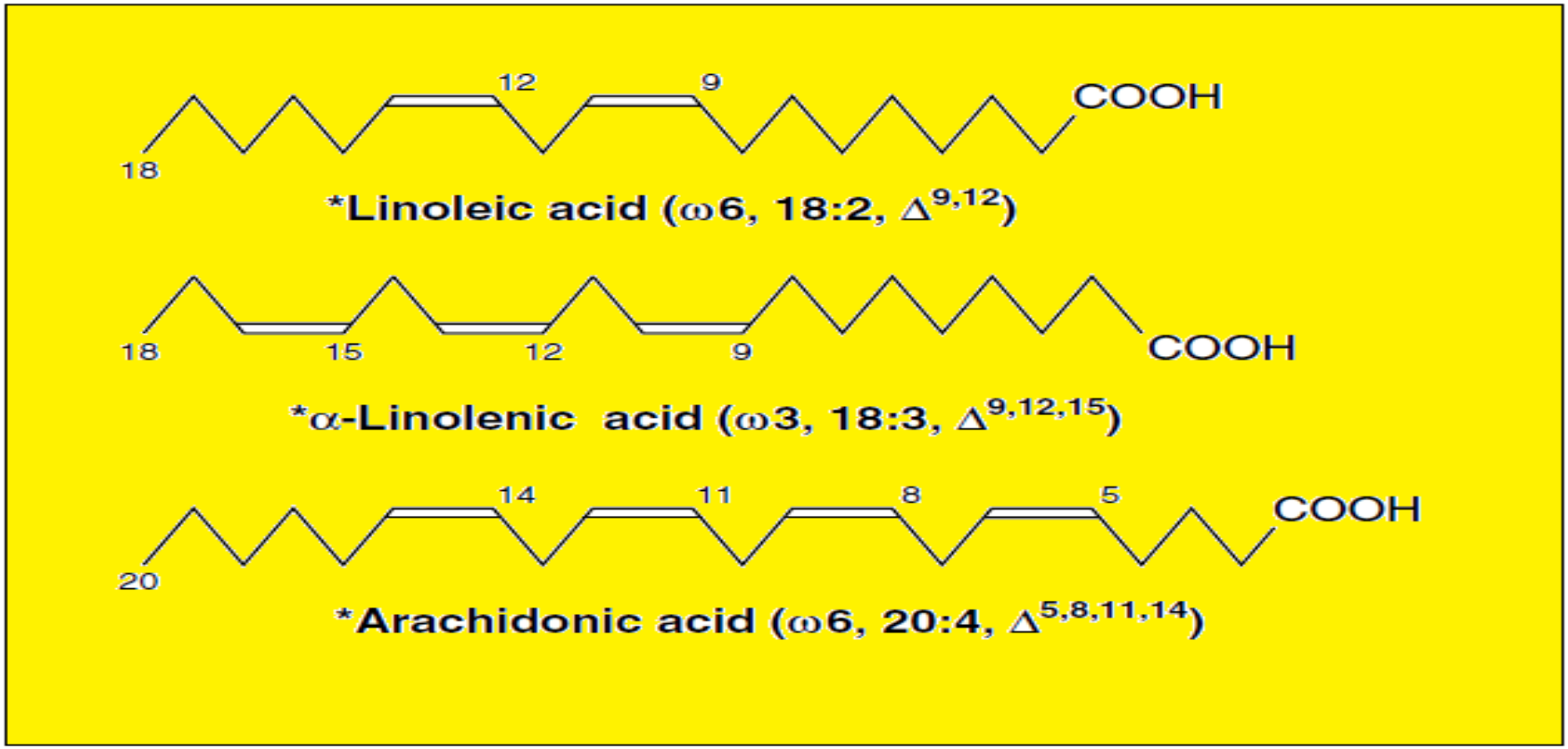
- Essential Fatty Acid
- Eicosanoids &Arachidonic acid
- Prostaglandins
- Overview of Prostaglandin Synthesis
- Biological functions of Eicosanoids

Membrane Lipids

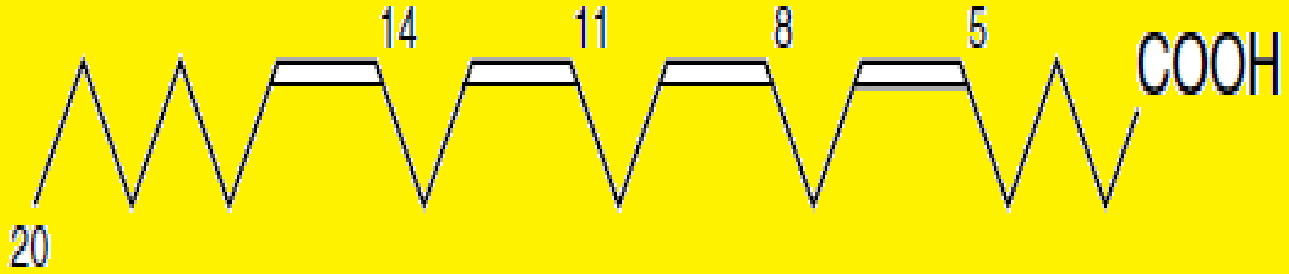



Phospholipid
(component of cell membrane)
Phospholipase A_2
Corticosteroids
(for example, cortisol)
Lysophospholipid
Arachidonic acid
5-Lipoxygenase
Cyclooxygenase 1
(COX-1, constitutive)
Cyclooxygenase 2
(COX-2, non-constitutive)
Cytokines, endotoxin, growth factors, tumor promotors
5-HPETE
(5-Hydroperoxyeicosatetraenoic acid)
Aspirin
(irreversible inhibitor)
Indomethacin
Phenylbutazone
(reversible inhibitors)
Corticosteroids
Selective COX-2 inhibitors
(for example, celecoxib, rofecoxib)
LTA_4 (Leukotriene A_4)
• Produced in leukocytes, platelets, mast cells, and heart and lung vascular tissues
PGG_2
Peroxidase, GSH
PGH_2
TXA_2 (Thromboxane A_2)
• Produced primarily by platelets
• Promotes platelet aggregation
• Vasoconstriction
• Mobilizes intracellular calcium
• Contraction of smooth muscle
Note opposing effects
$LTC_4 \rightarrow LTD_4 \rightarrow LTE_4$
• Contraction of smooth muscle
• Bronchoconstriction
• Vasoconstriction
• Increased vascular permeability
• Components of slow-reacting substance of anaphylaxis (SRS-A)
PGI_2 (Prostacyclin)
• Produced primarily by endothelium of vessels
• Vasodilation
• Inhibits platelet aggregation
LTB_4 (Leukotriene B_4)
• Increased chemotaxis of polymorphonuclear leukocytes
• Release of lysosomal enzymes
• Adhesion of white blood cells
$PGF_{2\alpha}$ (Prostaglandin $F_{2\alpha}$)
• Produced by most tissues
• Vasoconstriction
• Contraction of smooth muscle
• Stimulates uterine contractions
PGE_2 (Prostaglandin E_2)
• Produced by most tissues, especially kidney
• Vasodilation
• Relaxes smooth muscle
• Used to induce labor

Essential fatty acids



Eicosanoids: Arachidonic acid

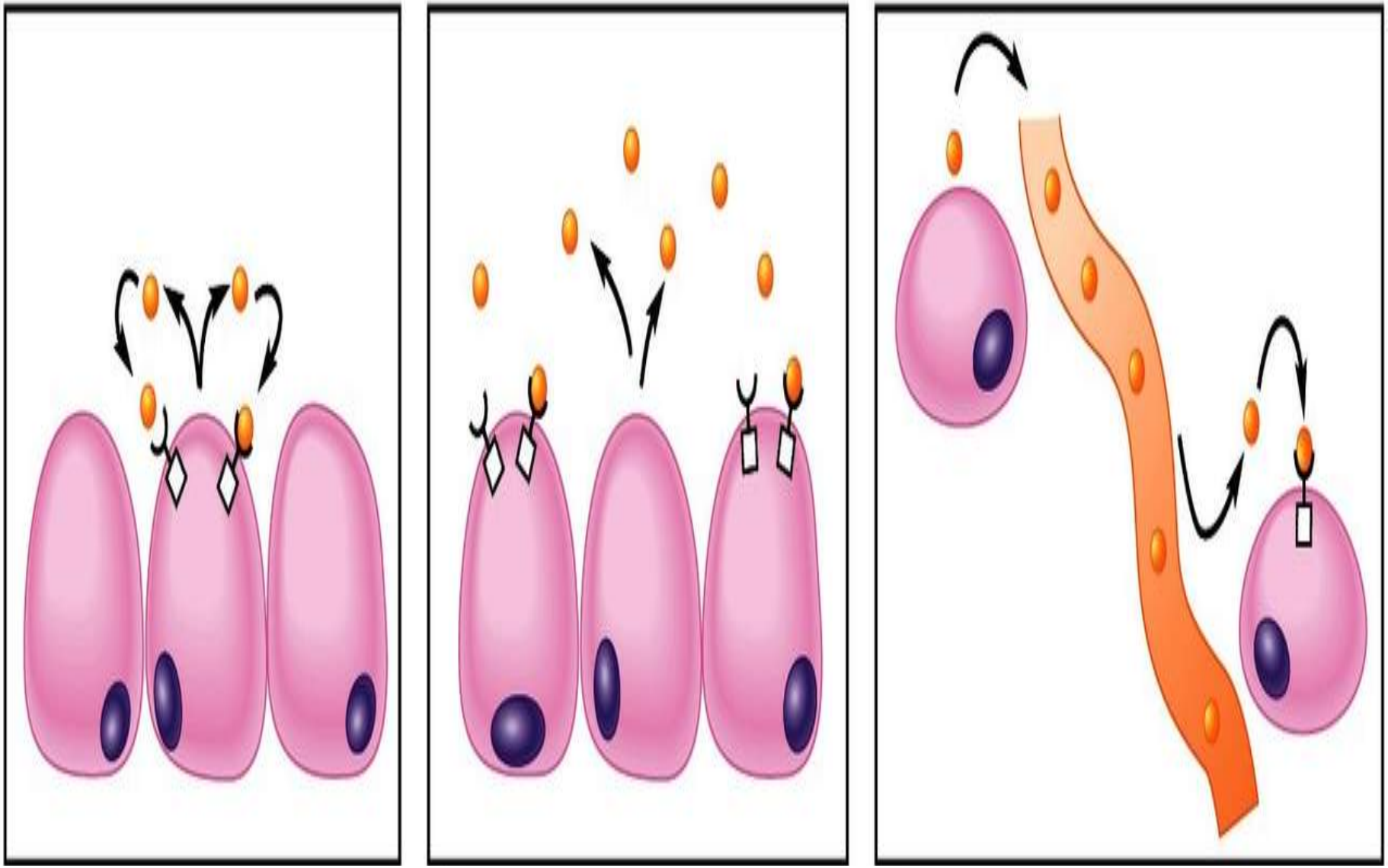


*Arachidonic acid (ω6, 20:4, $\Delta^{5,8,11,14}$)

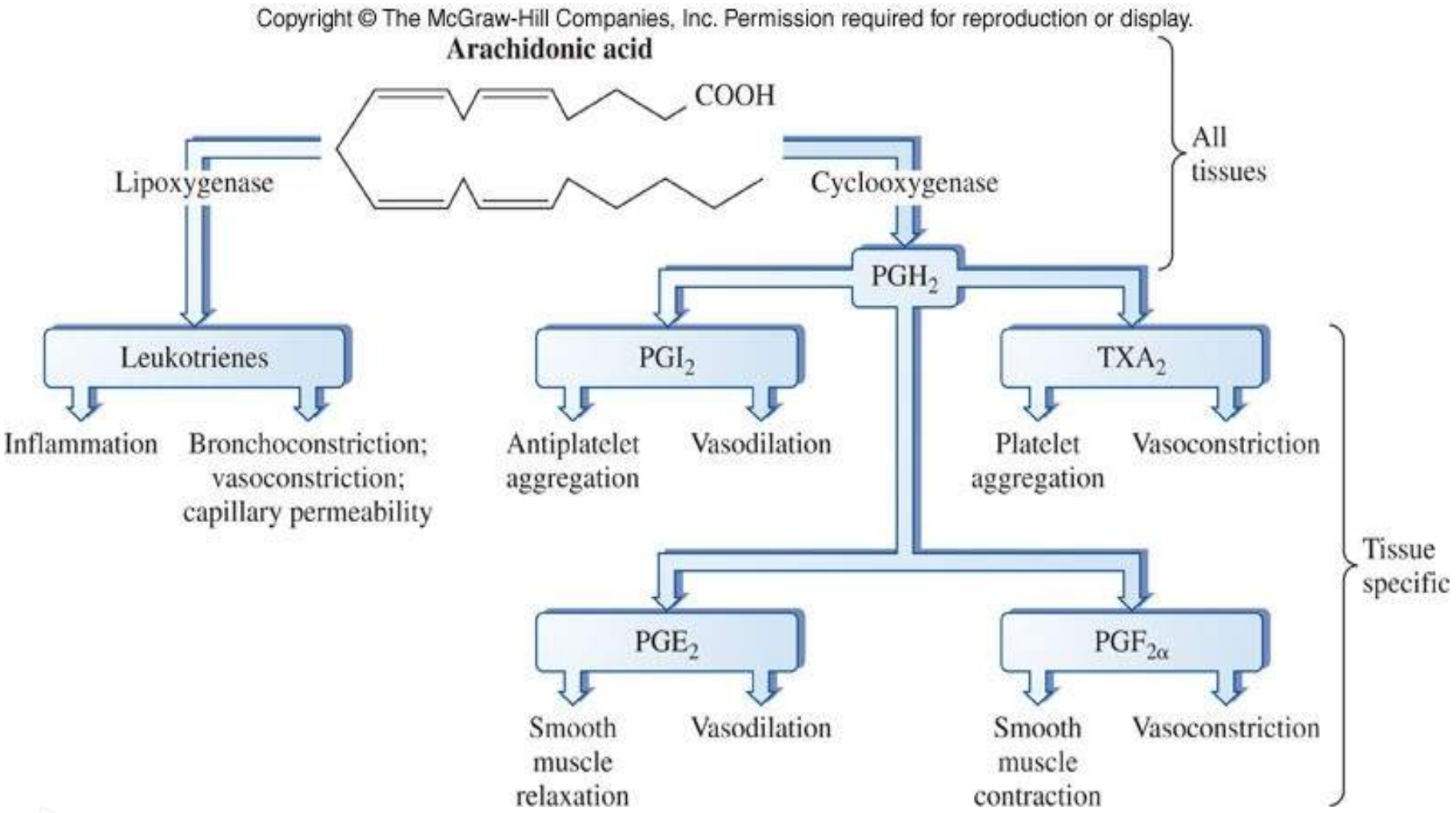
How many double bonds are in arachidonic acid?

4 = BOND

Prostaglandins



Overview of Prostaglandin Synthesis From Arachidonic Acid



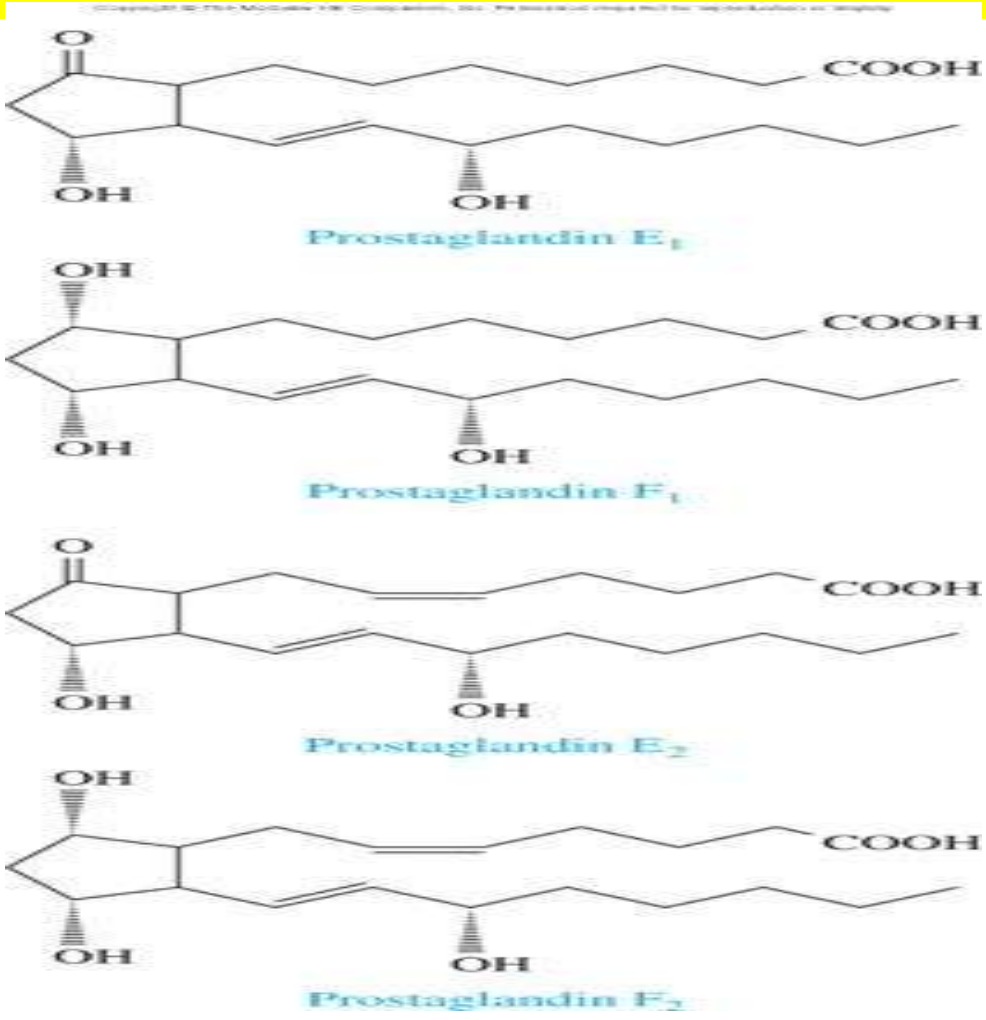
Biological Processes Regulated by Eicosanoids

1. Blood clotting
   - Thromboxane A_2 stimulates constriction of blood vessels and platelet aggregation
   - Prostacyclin dilates blood vessels and inhibits platelet aggregation
2. Inflammatory response
   - Prostaglandins mediate aspects of inflammatory response.
3. Reproductive system
   - Stimulation of smooth muscle by PGE_2

Biological Processes Regulated by Eicosanoids

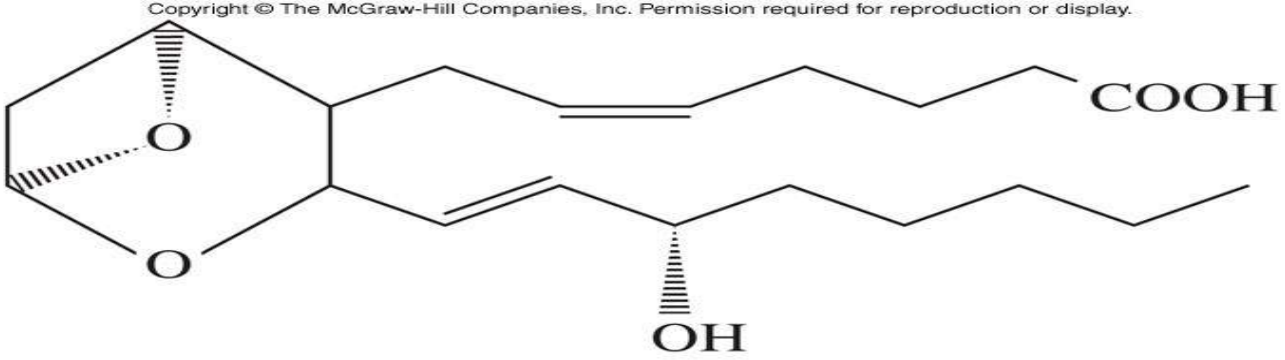
4. Gastrointestinal tract
   - Prostaglandins increase secretion of protective mucus
   - Inhibition of hormone-sensitive lipases
5. Kidneys
   - Prostaglandins dilate renal blood vessels
   - Results in increased water and electrolyte excretion
6. Respiratory tract
   - Leukotrienes promote the constriction of bronchi
   - Prostaglandins promote bronchodilation

Structures of Four Prostaglandins

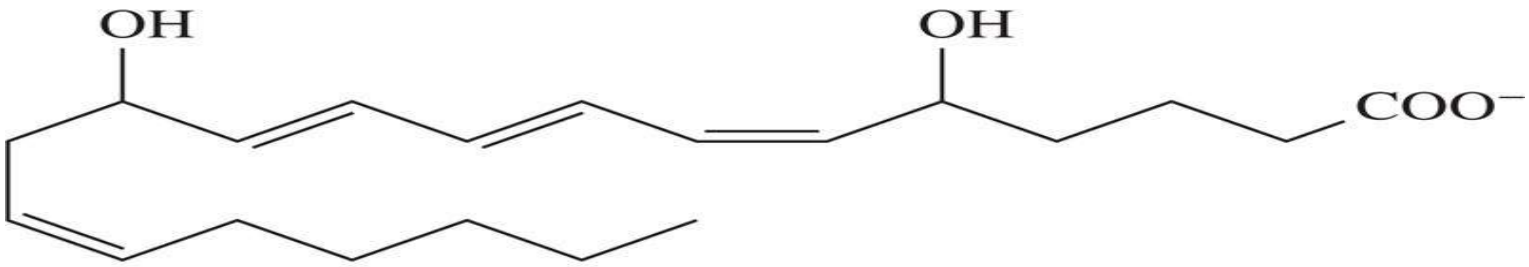


Thromboxane and Leukotriene Structure

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Thromboxane A_2

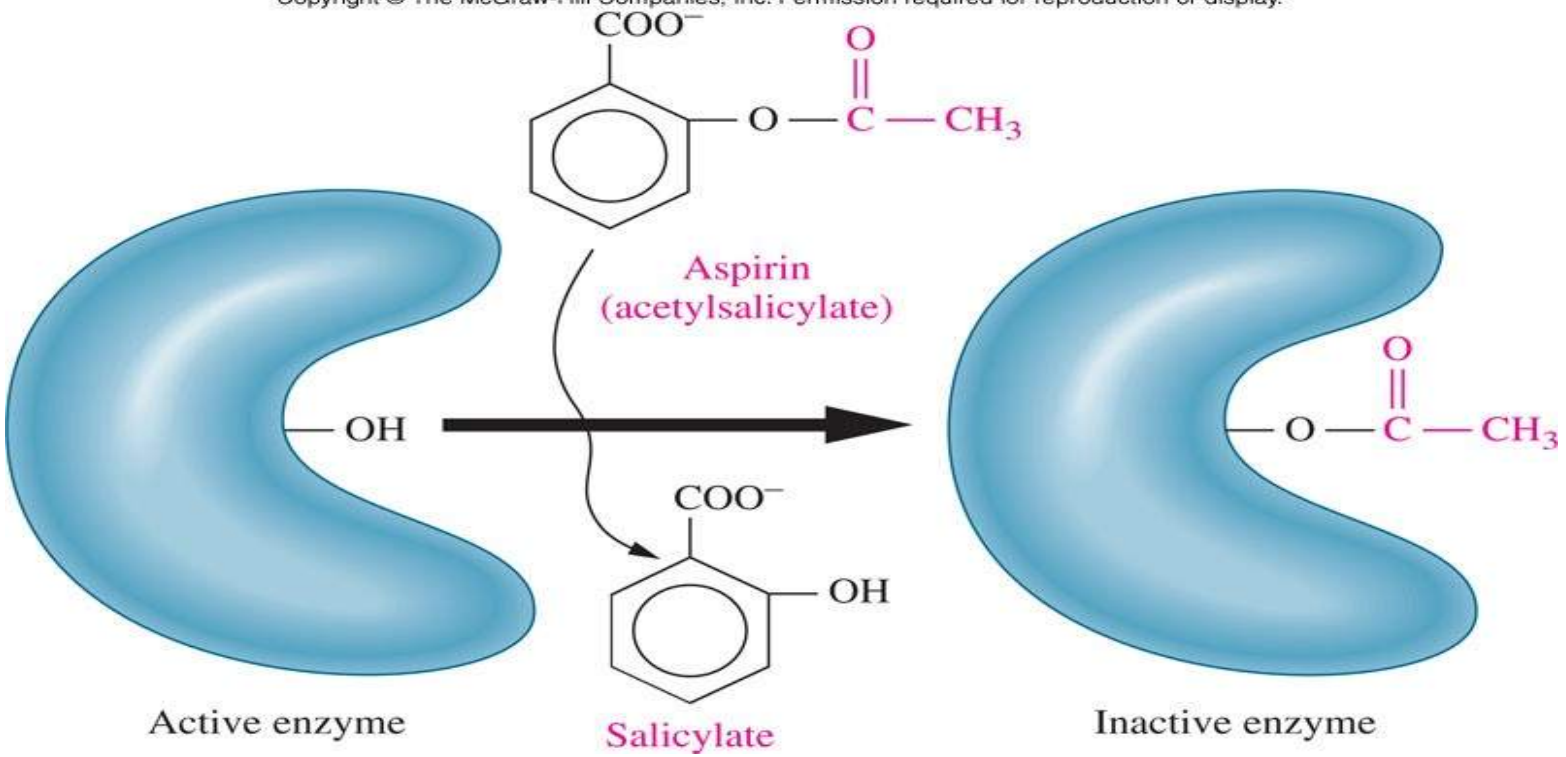


Leukotriene B_4

Aspirin and Prostaglandins

Aspirin inhibits prostaglandin synthesis by acetylating cyclooxygenase, an enzyme necessary for prostaglandin synthesis

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17.2 Fatty Acids

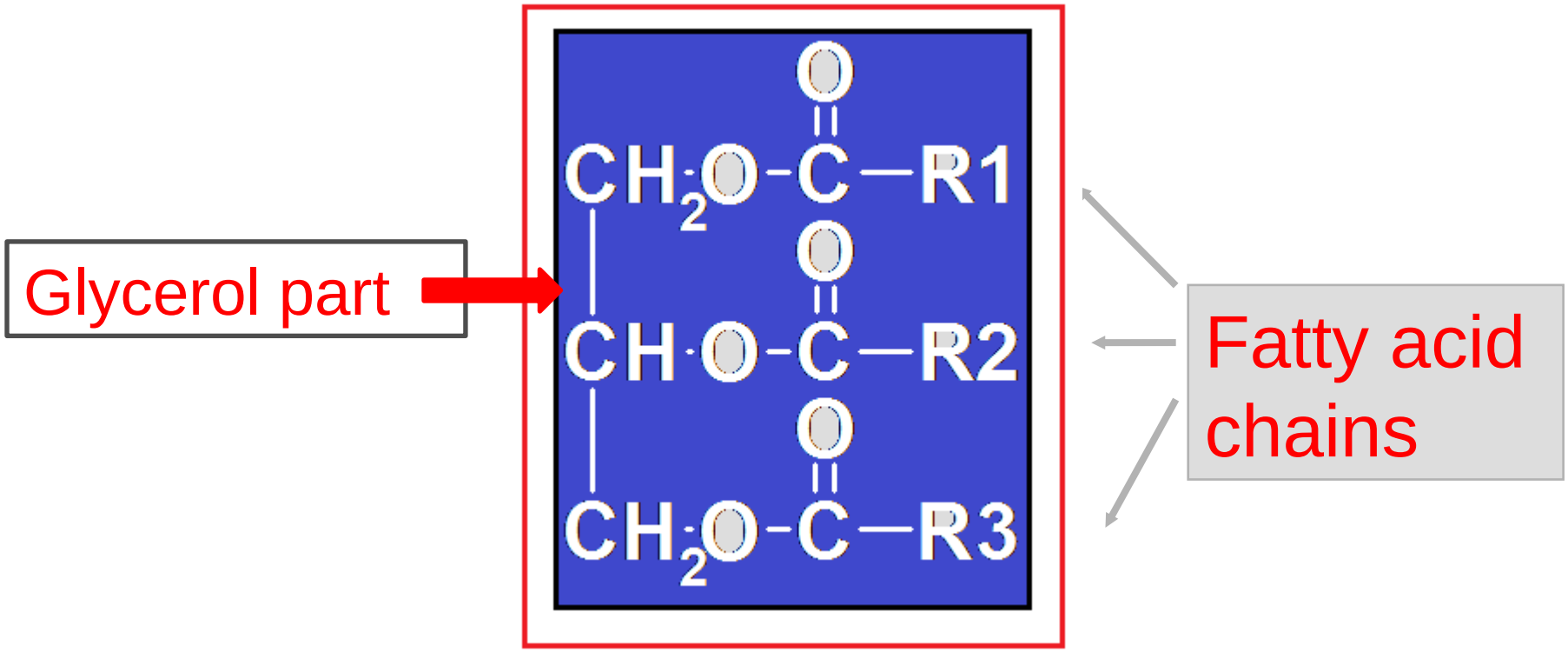
Derived Lipids
GLYCERIDES

Glycerides

- Glycerides are lipid esters
  - Alcohol group of glycerol form an ester with a fatty acid
  - Esterification may occur at one, two, or all three alcohol positions producing:
    - Monoglyceride
    - Diglyceride
    - Triglyceride
      - A neutral triacylglycerol or a triglyceride
      - Triglycerides serve as energy storage in adipose cells

Triglycerides

- Glycerides are lipid esters
- A triglyceride places fatty acid chains at each alcohol group of the glycerol



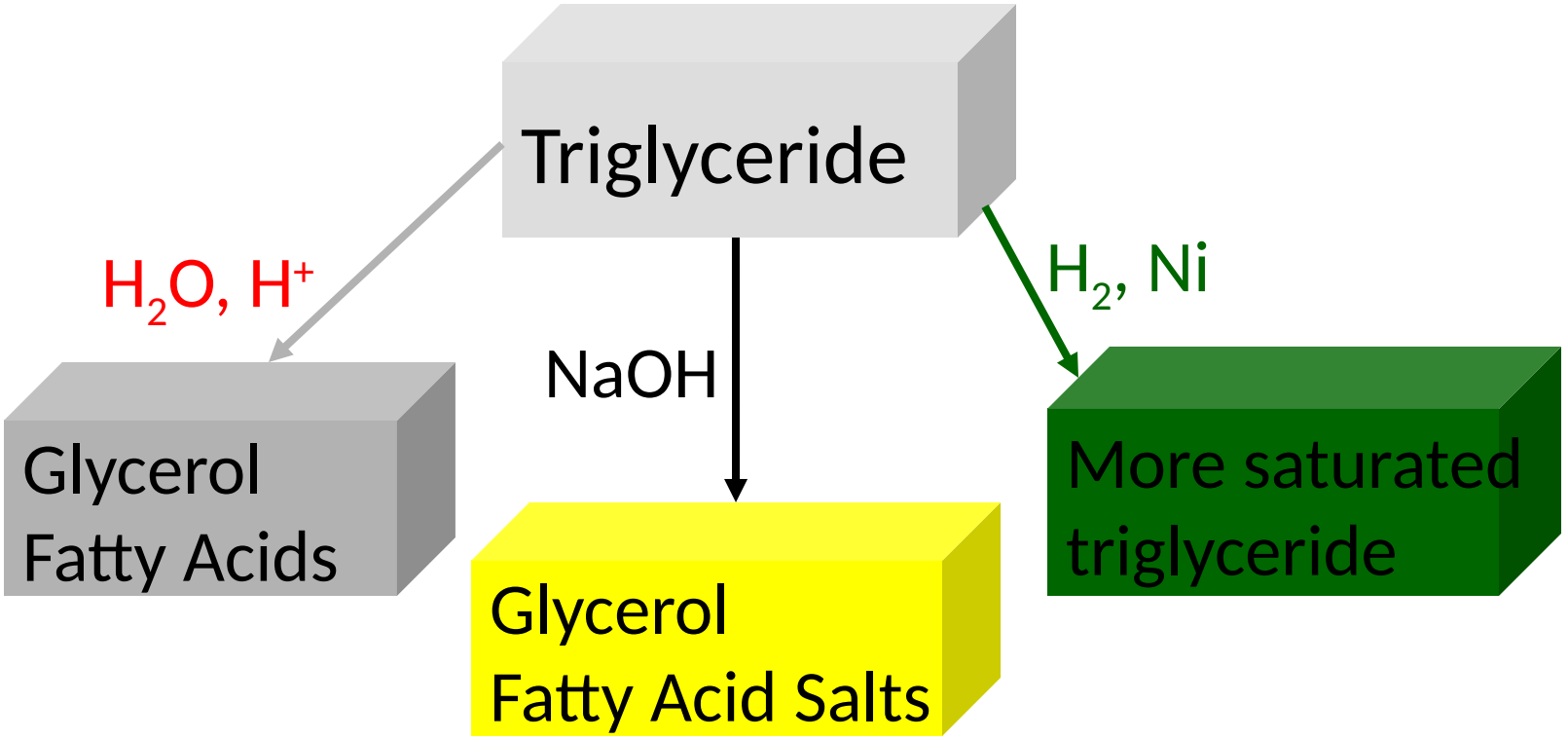
Chemical Properties

Triglycerides have following chemical properties as they are composed of these two groups:

- Hydrolysis: produces the fatty acids and glycerol, a reverse of formation
- Hydrogenation: saturates the double bonds.
- Saponification: formation of Salt with Strong Base

Triglyceride Reactions

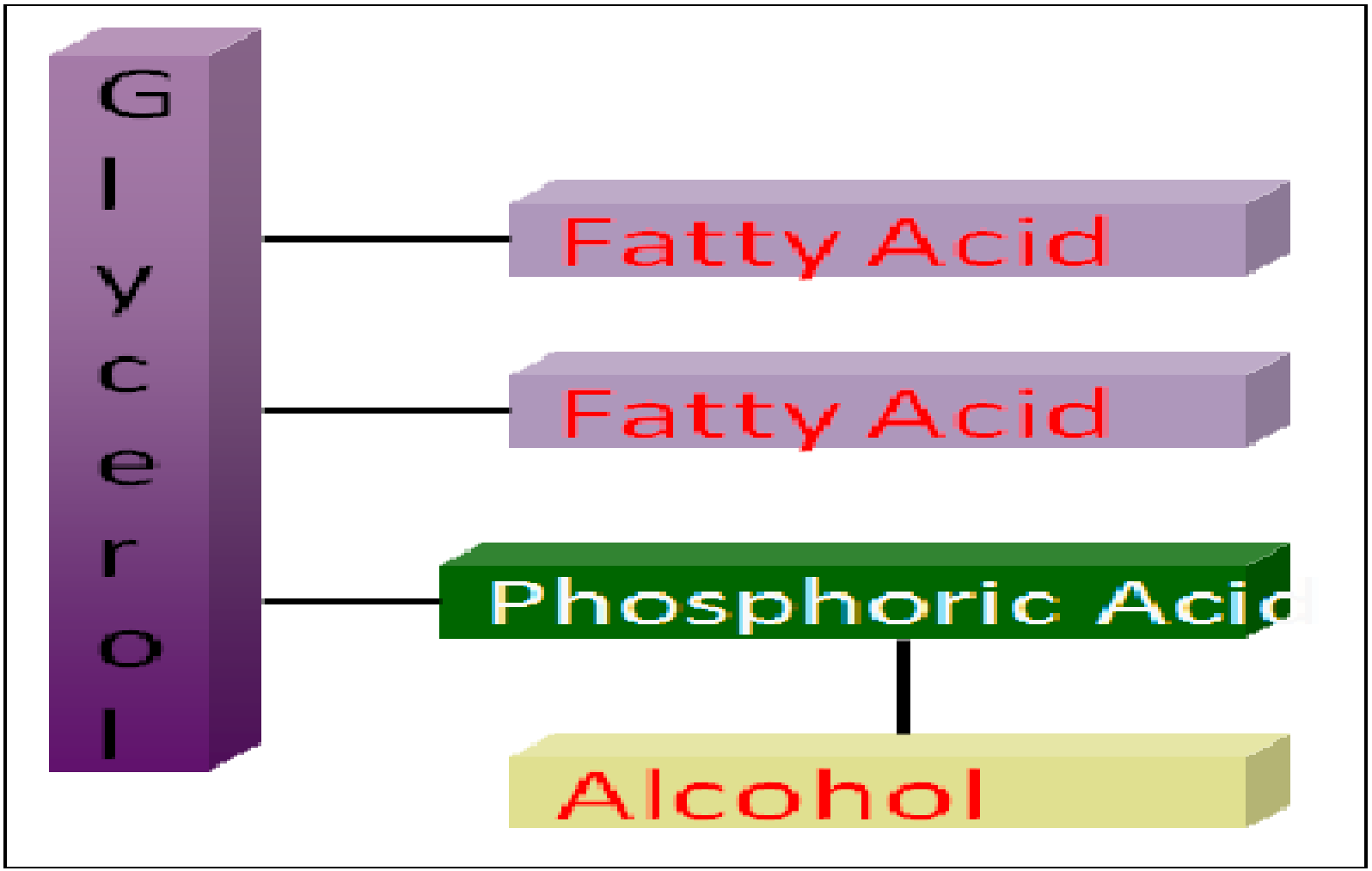
- Triglycerides undergo three basic reactions
- These reactions are identical to those studied in carboxylic acids



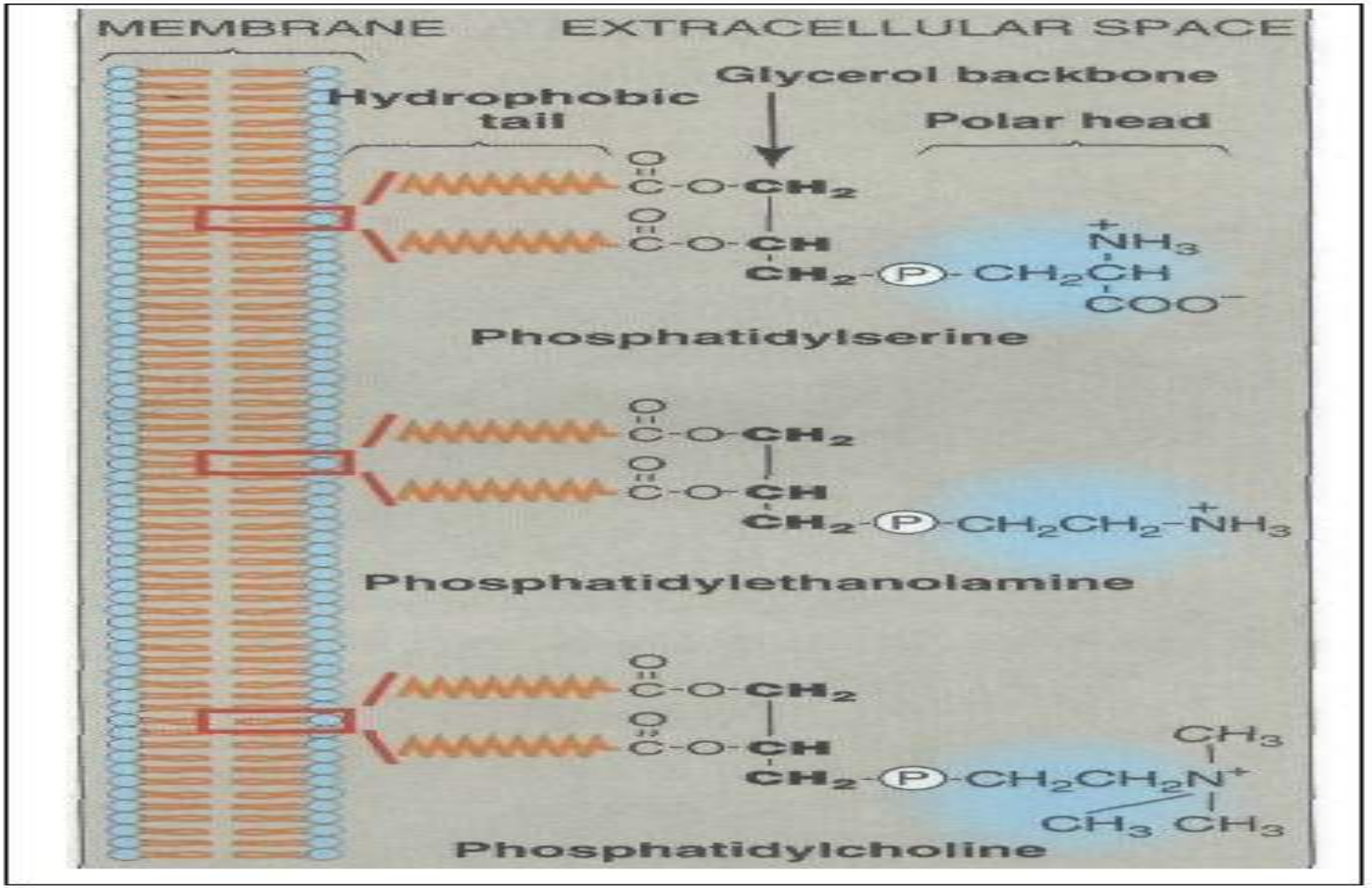
Fats and Oils

- Triglycerides or triacylglycerols
  - Fats are a combination of glycerol and the fatty acids
- Fats mainly come from animals and are solid at room temperature
- Oils mainly come from plants, and are liquid at room temperature

Phosphoglycerides



Types of Phosphoglycerides



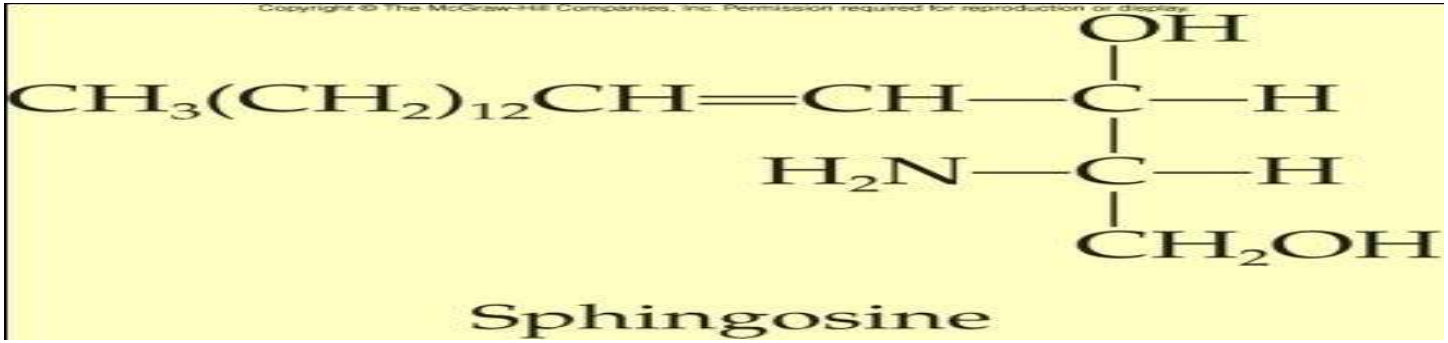
Respiratory Distress Syndrome(RDS) & Role of Surfactant

- Surfactant serves to decrease the surface tension of this fluid layer, reducing the pressure needed to reinflate alveoli, thereby preventing alveolar collapse.
- **Respiratory distress syndrome** (RDS) in pre-term infants is associated with insufficient surfactant production, and is a significant cause of all neonatal deaths.
- An L/S ratio greater than 2.2 is considered mature

Nonglyceride Lipids

Sphingolipids

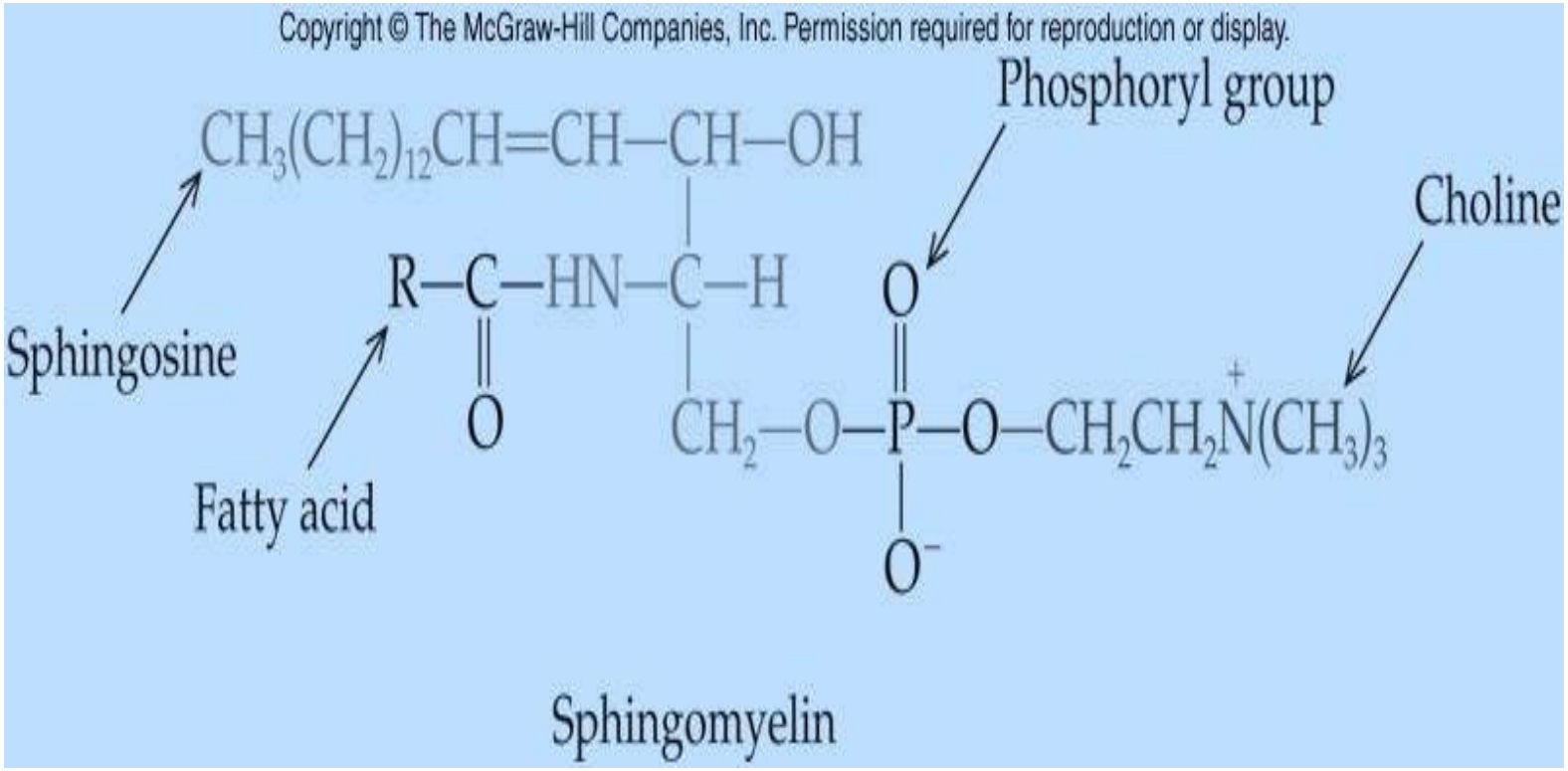
- These lipids are based on sphingosine
- A long-chain, unsaturated nitrogen containng alcohol [$C_{18}H_{37}NO_2$]



- Structural component of cellular membranes
- Major categories
  - Sphingomyelins
  - Glycosphingolipids

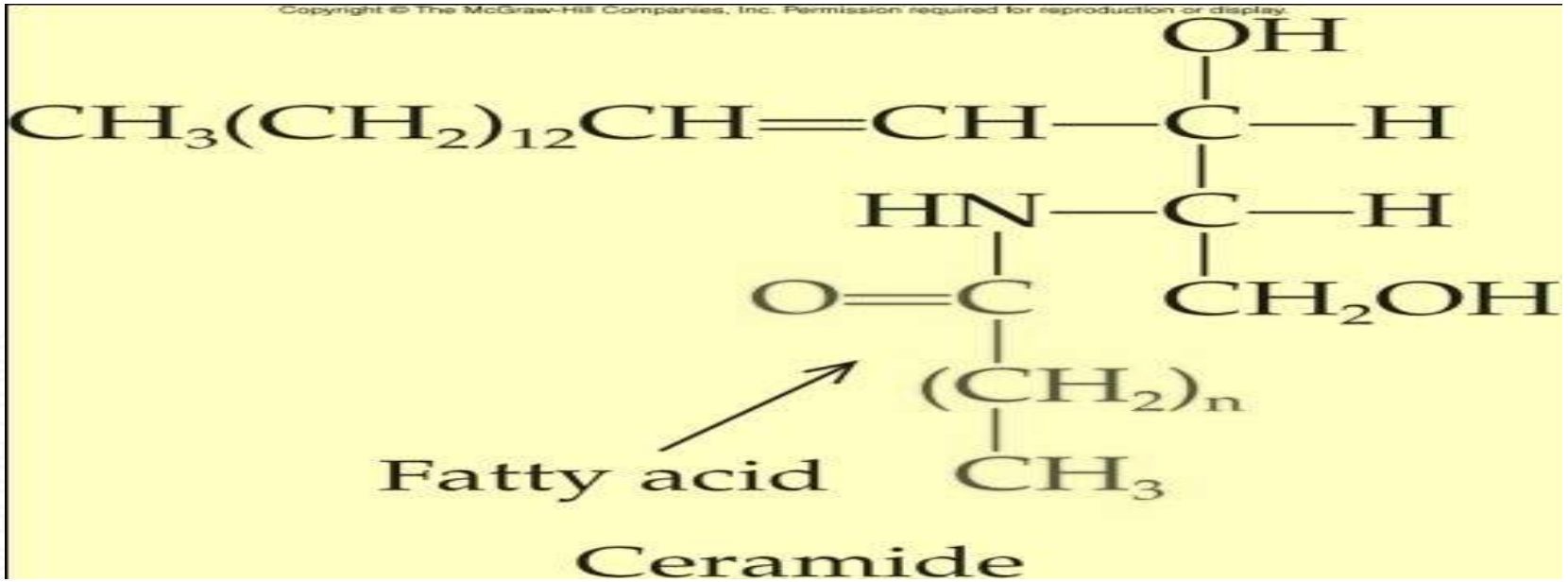
Sphingomyelins

- Sphingomyelin is a sphingosine-based phospholipid
  - Structural lipid of nerve cell membranes and Myelin sheath.



Glycosphingolipids

- It is type of Glycosphingolipids and is built on a ceramide.
- A ceramide is a lipid molecule composed of the amino acids sphingosine and a fatty acid



17.4 Nonglycerol lipids

Type of Glycosphingolipids

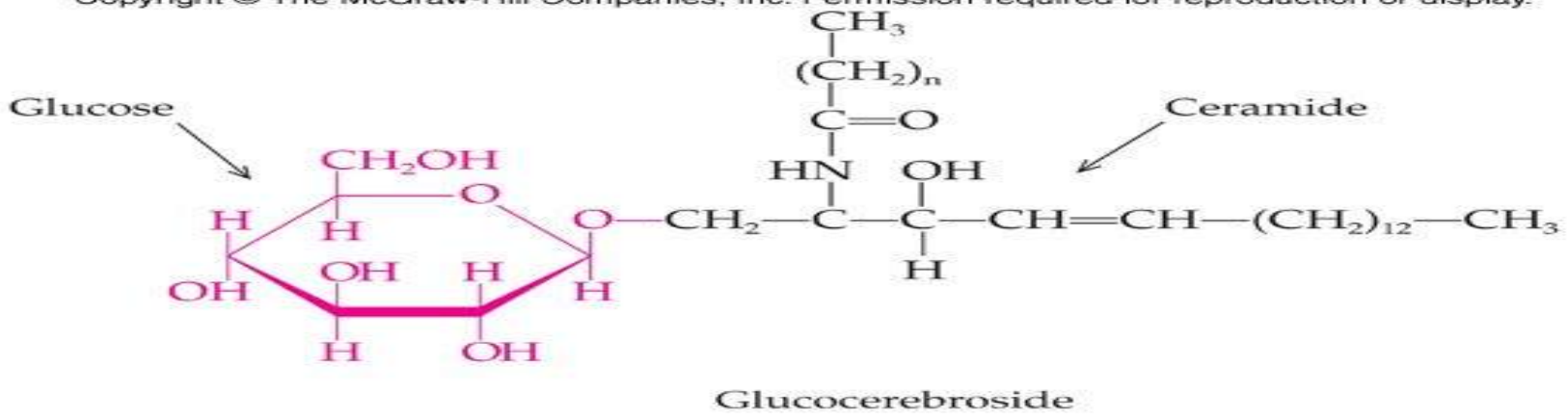
- **Neutral glycosphingolipids**
  - Glucocerebroside
  - Galactocerebroside
- Cerebrosidesare found predominantly in the brain and peripheral nervous tissue,with high concentrations in the myelin sheath.
- **Acidic glycosphingolipids**
- Gangliosides found primarily in the ganglion cells of the central nervous system, particularly at the nerve endings.
- Sulfatides: Sulfatides are found predominantly in nerve tissue and kidney.

Neutral Glycosphingolipids

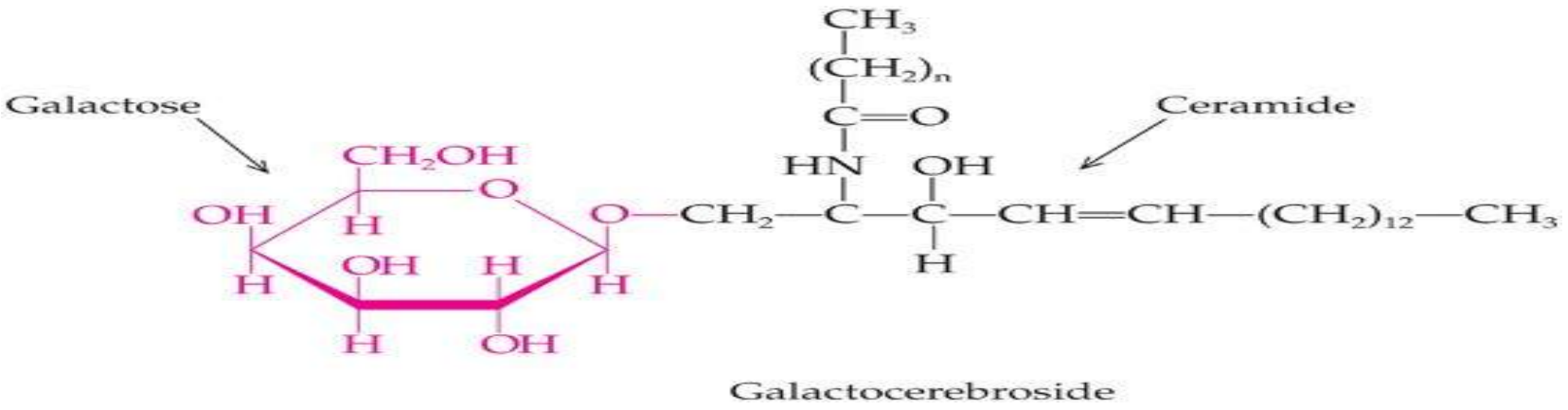
The simplest neutral glycosphingolipids are the **cerebrosides.**

- Glucocerebroside
- Galactocerebroside

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Glucocerebroside



Galactocerebroside

Cis and Trans configuration

- A type of geometric isomerism
- “Orientation of atoms or groups around the axes of double bonds”

Cis and Trans configuration

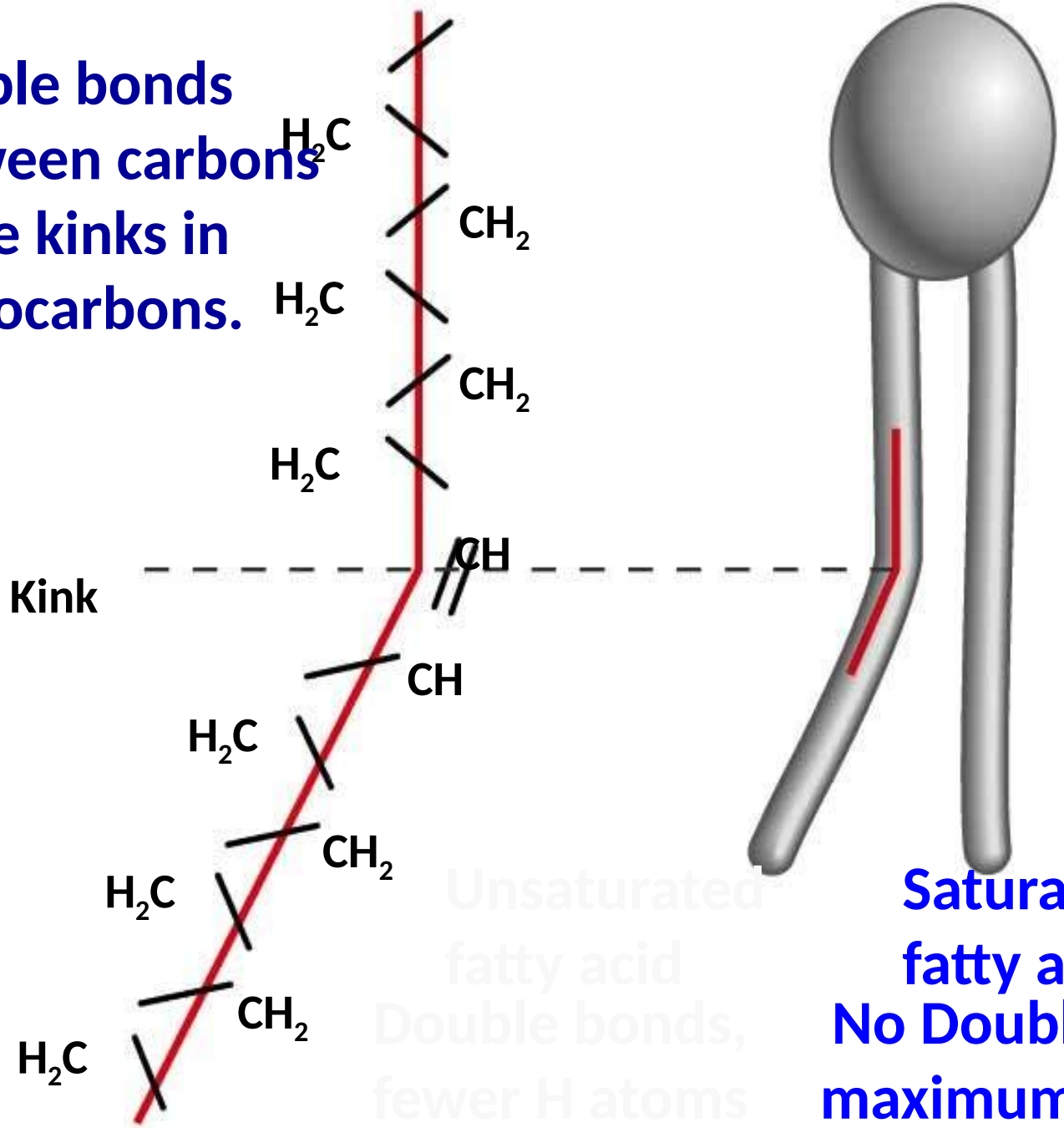
Cis configuration

- Atoms or groups are on the same side of the double bond

Trans configuration

- Atoms or groups are on the opposite side of the double bond

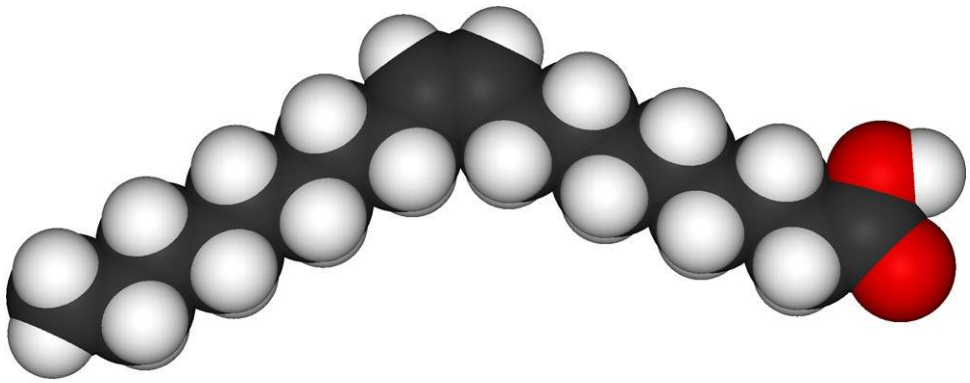
Double bonds between carbons cause kinks in hydrocarbons.



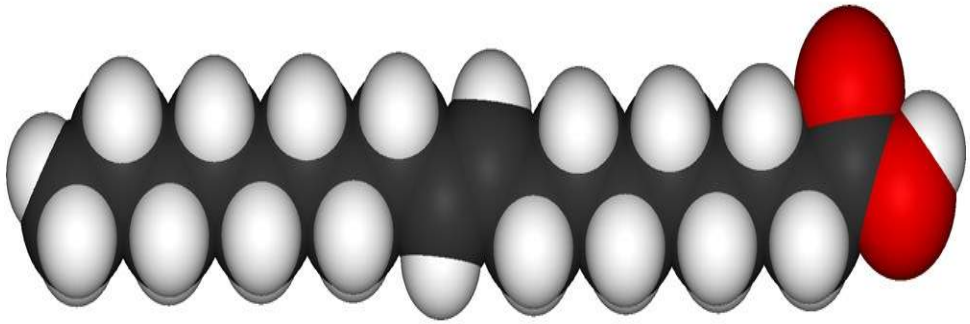
Saturated fatty acid

No Double bonds, maximum H atoms

- **Oleic acid** is a ***cis*** unsaturated FA that comprises 55–80% of olive oil.

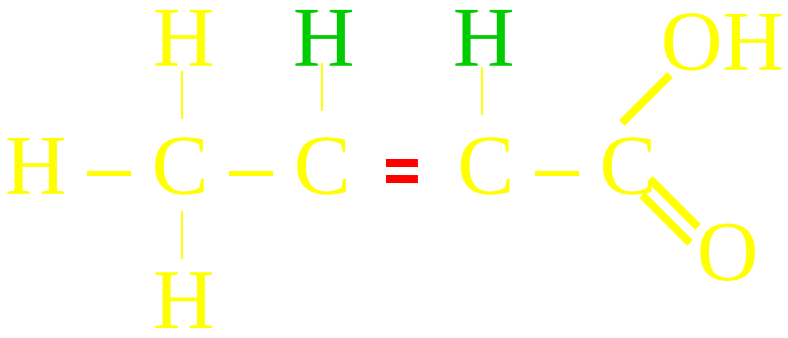


Elaidic acid is a ***trans*** unsaturated FA often found in partially hydrogenated vegetable oils.

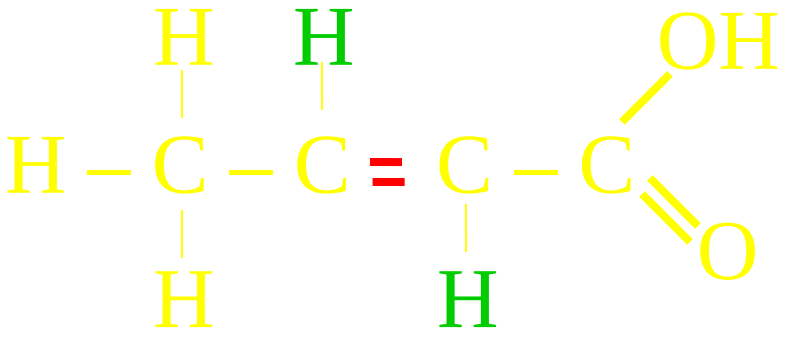


Most fatty acids are CIS configuration

CIS configuration results in chain kinking

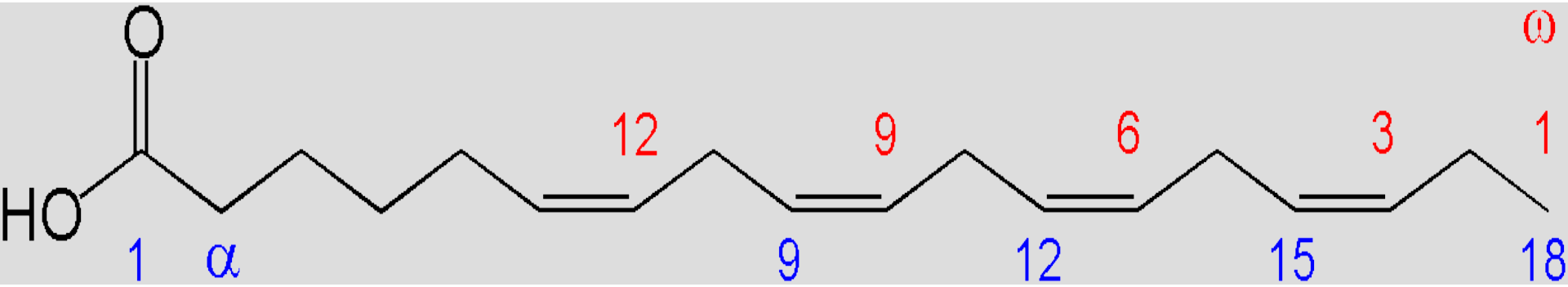


CIS

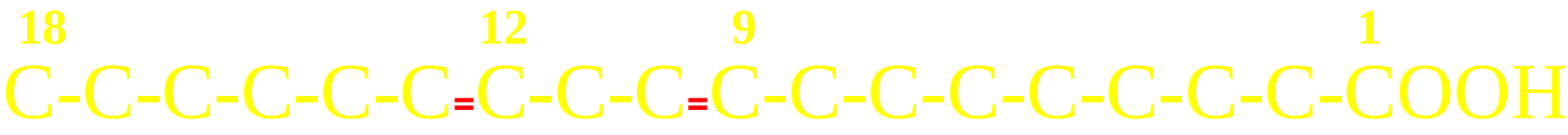


TRANS

Omega (ω) Classification



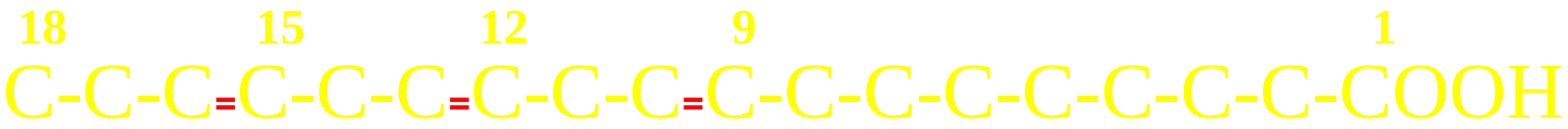
Omega (ω) Classification



18 : 2 ($\Delta^{9,12}$) **ω= 6**

Note: we always count from the Omega carbon. (Linoleic acid)

Omega (ω) Classification



18 : 3 ($\Delta^{9,12,15}$) **ω= 3**

Note: we always count from the Omega carbon. (Linolenic acid)

Fatty acid composition of TAG in:

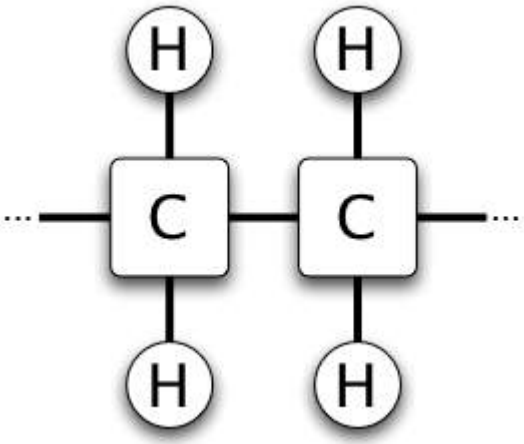
	C_{4-14} Short chain saturated	$C_{16/18}$ Long chain saturated	$C_{16/18}$ unsaturated
OLIVE OIL	5	15	80
BUTTER	20	40	40
LARD	5	50	45

What are trans fatty acids and where do they come from?

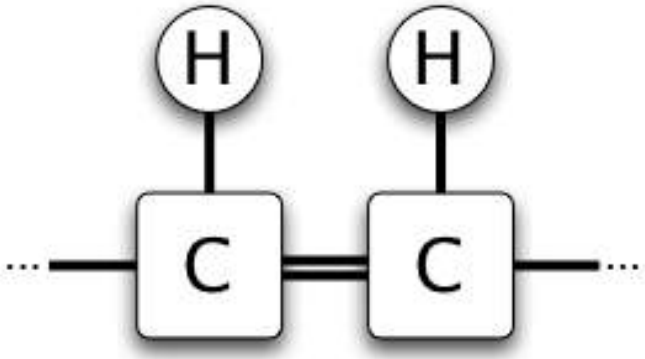
During hydrogenation, some hydrogen atoms move to the opposite side of a newly formed double bond (trans fatty acids).

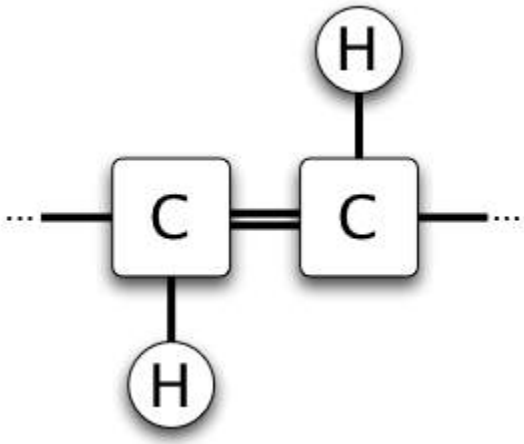
Trans fatty acids (trans fats) contribute to the stability of hydrogenated fat thus preventing rancidity.

Saturated fat



Trans-unsaturated FA





Cis-unsaturated FA

- Trans double bonds are also formed during the hydrogenation of either vegetable or fish oils.
- French fries, donuts, cookies, chips and other snack foods are high in trans fatty acids.
- Nearly all fried or baked goods have some trans fats.

Should we worry about Trans Fatty Acids?

Trans fatty acids tend to raise blood cholesterol levels, as compared with Monounsaturated and Polyunsaturated fatty acids in the diet.

Importance EFA

- Structural component of tissues
- Synthesis of prostaglandins and leukotrienes
- Lowering of serum cholesterol level
- Role in vision (Docosahexenoic acid in retina enhances the electrical response of the photoreceptor to light).
- Prevent fatty liver.
- Structural component of gonads and mitochondrial membranes

Deficiency of EFA

- In adults on normal diets, no signs of EFA deficiencies have been reported.
- In infants receiving formula diets low in fat
- Patients getting intravenous nutrition for long periods show deficiency symptoms.
- Skin lesions and impairment of lipid transport.

